

Toward Quasi-Polynomial Unknot Recognition

Presentation complexity, early first-jet certificates,
and streaming boundary responses

Research continuation prepared for Vladimir Reshetnikov

8 October 2026

Abstract

We develop three exact improvements to the ProveIt unknot-recognition program, separating complete parameterized decision theorems from optional implementation experiments. First, a presentation with r generators and total explicit relator length N yields deterministic recognition in $\text{poly}(N + r)(2 + N/r)^{O(r)}$ bit operations, after verified provenance is included. A straight-line word presentation gives a constant-degree formulation with cost $\text{poly}(S + r + s)2^{O(r+s)}$; a cut through its word circuit gives $\text{poly}(S + r + s)(d + 2)^{O(r+t)}$. The proofs account for dense polynomial construction, coefficient height, inverse letters, conjugate meridians, and degenerate gauge choices.

Second, two binary matrix ranks compute a one-matching complex's scalar survivors and first-jet closure multiplier before cobordism elimination. A rigidity theorem at multiplier one proves equality of total Khovanov rank under every classical link continuation. A separate flat-origin constraint explains why the stronger algebraic interval compression is absent from genuine minimal complexes in the tested corpus.

Third, a reverse filtration of cut faces compiles all suffix determinant responses in $O(n(W+1)^2)$ field operations, retaining singular interior directions and handling characteristic two. Controlled experiments show substantial response-setup improvements, mixed early-observer performance, and useful small-instance exact algebraic decisions. All inconclusive solver outcomes and unfavorable timings are retained.

The work supplies proved sufficient quasi-polynomial regimes and explicit obstructions to universal whole-group compression. It does not establish a quasi-polynomial bound for the full maintained recognizer. We conclude with a prioritized research program, concrete proof obligations, a patch against a pinned repository revision, and reproducibility artifacts.

Source baseline: `1be2abc8cc743c1a3b5cb7fdfeb650697f8e2d5b`.

Repository: [ProveIt / Topology / UnknotRecognition](#).

Companion archive contains source, exact inputs, raw measurements, tests, integration patch, and build instructions.

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1 Scope, status, and the three costs being reduced

The objective is a mathematically justified acceleration of the ProveIt unknot-recognition algorithm, with quasi-polynomial asymptotic complexity as the research target. The repository already contains a substantial collection of exact structural certificates, compressed group operations, and relative Khovanov scanners. A useful continuation must identify which expensive operations it actually avoids and which input-size parameters remain uncontrolled. This report develops three such improvements, and includes the code and negative experimental results needed to evaluate them independently.

Definition 1.1 (Input and target). The primary input is a validated spherical planar diagram of a tame knot with n crossings and one component. Crossing and edge labels have polynomially bounded encoding length after normalization. General quasi-polynomial time means $\exp((\log(n+2))^{O(1)})$; several sufficient conditions in this report yield the more specific bound $\exp(O((\log(n+2))^2)) = n^{O(\log n)}$. Bounds for a supplied compressed presentation explicitly include its description length and provenance.

Three different costs occur. The first is the number of real variables and the polynomial degree in an exact representation query. The second is cobordism-valued elimination performed before a decision certificate becomes visible. The third is repeatedly eliminating the same graph interior while preparing boundary observations at successive stages. Reducing any one of these can be useful. None alone bounds the total number of intermediate relative-complex objects or guarantees a short geometric decomposition.

Result	Proved guarantee	Remaining condition or limitation
Explicit knot-group presentation	$\text{poly}(N+r)(2+N/r)^{O(r)}$ bit operations	Construct and verify the presentation; N counts explicitly written letters.
Word-circuit cut	$\text{poly}(S+r+s)(d+2)^{O(r+t)}$ bit operations	Pay for retained gates, formal degree, dense compilation, and provenance.
Pre-elimination first jet	$t = N - 2 \text{rank } A$ and $\kappa = N - \text{rank} \begin{pmatrix} A & 0 \\ B & A \end{pmatrix}$	Whole one-matching summand; general multiplier needs a knot completion.
Multiplier-one rigidity	One matching preserves total rank under every classical link continuation	Preserves a decision quantity; original quantum grading is not retained.
All-suffix responses	$O(n(W+1)^2)$ field operations for every suffix	Prescribed order; query count and relative-complex growth are separate.
Checkpoint scan	$\text{poly}(n)2^{O(g+w)}$ for its actual maximum gap and frontier	No universal polylogarithmic bound on $g+w$ is proved.

Table 1: The theorem boundaries. Letters reused inside individual sections are defined locally; in the presentation rows r counts generators and N counts relator letters, while in the first-jet row N counts complex objects.

1.1 What is established and what is still research

The presentation theorems are exact decision theorems using established existential real algorithms. The delivered compiler emits exact integer polynomials, but its optional practical solver is NLSAT through Z3; the theoretical running-time bound is not attributed to that solver. The first-jet scanner is a complete rank-decision method when its underlying exact computation is allowed to finish, with a bounded optional observer and an unchanged mathematical fallback. The streaming-response

module is a subroutine with a proved arithmetic bound, and is not installed in the default recognition pipeline.

The development is based on the pinned revision `1be2abc8cc743c1a3b5cb7fdfeb650697f8e2d5b` of the ProveIt repository [23]. Prior reports establish the binary Jones contraction, scalar residue and radical analysis, first-jet knot-completion multiplier, and fixed-stage boundary responses. The new work is the quantitative presentation compilation, the placement of the first-jet test before elimination together with its multiplier-one link extension and flat-origin constraint, and reuse of boundary elimination across all changing suffixes. Established ingredients receive attribution; the fact that a combination is proved here does not establish historical priority for every ingredient or parameterized consequence.

Current literature and the meaning of the target. The present work does not claim a new general quasipolynomial algorithm. Lackenby’s March 2021 author slides announce a running time $2^{O((\log n)^3)} = n^{O((\log n)^2)}$ and outline a hierarchy approach [11]. We use that source’s displayed bound. His July 2026 preprint proves a new hierarchy algorithm for incompressibility, with unknot recognition as a consequence, and explicitly leaves possible speed-ups to future work [12]. It should therefore not be cited as a completed proof of the announced quasipolynomial bound. The present algebraic sufficient condition targets $n^{O(\log n)}$ on a specified residual class, under explicitly stated and verified compression hypotheses. These are different statements.

A September 2026 preprint by Musick claims a polynomial-time procedure for producing locally minimal bridge presentations and, consequently, unknot recognition in P [17]. Version 2 is dated 20 September 2026. That claim has not been independently established by this study and none of the results or code here depends on it. Recording its existence avoids the inaccurate claim that no polynomial-time preprint has appeared; treating it as an unverified recent claim avoids replacing proof checking with an abstract-level assertion. An independent audit would have to establish the completeness of the permitted presentation moves and the polynomial bounds on their compressed execution, as well as the correctness of the initial diagram conversion.

What the hierarchy bounds suggest. Lackenby’s 2026 Proposition 9.1 bounds the number of main iterations by $L(g + 1)^L$, where L controls hierarchy length and g controls the relevant pattern complexity. The later sections develop efficient hierarchy certificates and compressed cutting operations [12]. Purely as an implication of that bound, polynomial g and $L = O(\log n)$ would yield $n^{O(\log n)}$ iterations, while $L = O((\log n)^2)$ yields $n^{O((\log n)^2)}$. Counting iterations alone still requires a bound on the bit cost of each iteration and on the construction of the surfaces used. The 2021 slides describe polynomial genus, compressed hierarchies, multisurfaces, and Heegaard/Cheeger-region arguments as the components of that route [11]. A rigorous implementation project would need exact data formats, verified transition rules, and complete size and bit-complexity bounds for those components.

1.2 A stronger sufficient condition is useful even with obstructions

For a word circuit, let r count original generators, let t count retained gates, and let d bound the formal degrees after cutting. The explicit target

$$(r + t) \log(d + 2) = O((\log n)^2)$$

turns an informal request to “compress the group” into a testable theorem statement. It must be accompanied by polynomial construction and verification costs. Connected sums of trefoils, and a

prime pretzel family, have group rank growing linearly with crossing count. Thus this condition cannot hold universally for presentations of the entire knot group. The same knots have inexpensive coloring obstructions, so they do not refute a portfolio that first removes such cases. A meaningful residual-class theorem must state those filters and decompositions exactly.

Likewise, the first-jet scan supplies an explicit target $g + w = O(\log^2 n)$ on the executed order, including temporary crossing allocation and the last segment. The streaming response theorem reduces the cost of observing such an order without proving that the order exists. These formulations make a future quasi-polynomial claim susceptible to proof checking rather than extrapolation from favorable timings.

2 The maintained algorithm and its unresolved size parameters

2.1 The exact foundation

The maintained program validates diagrams before applying inexpensive structural tests, diagram simplification, invariant obstructions, and optional certificate searches. It has specialized recognition routines for several structured inputs and an exact Khovanov fallback. The group stage supports explicit and straight-line words, with independent replay of its existing certificate formats. Optional normal-surface calls have a separate external-engine trust boundary. These are important existing capabilities; this report does not reintroduce them as new results [23].

The homological fallback uses Bar-Natan’s local tangle formalism and cancellation [1, 2]. A crossing extends a relative complex. Closed circles are delooped, scalar isomorphisms are cancelled, and dot polynomials encode the remaining cobordisms. At final closure the total characteristic-two Khovanov rank decides the unknot, using Kronheimer–Mrowka’s detection theorem and the characteristic-two reduced splitting [10, 24]. Section 4 states the exact rank convention used by the new decision scanner.

An invariant obstruction is logically asymmetric. A determinant or Jones value incompatible with the unknot proves knottedness, but a compatible value does not generally prove triviality. A verified reduction to the empty diagram, a replayed group certificate, or a completed detecting homology computation can certify the unknot. Resource exhaustion produces an unresolved status. This asymmetry remains part of every new interface.

2.2 Why width alone does not bound the scanner

Let w_s be the actual frontier at stage s , let M_s be its number of live objects, and let L_s be the explicit length of its coefficient data. A width bound controls the kinds of boundary matchings and the available dot coordinates. It does not bound the multiplicity of each matching or the number of homological copies. Even one fixed matching supports free complexes with arbitrarily many objects. A bounded algebra dimension therefore does not imply a bounded differential matrix.

Moreover, Gaussian elimination can temporarily construct large coefficient compositions even when its final scalar survivor count is small. A correct cost model must include each crossing’s complete allocation, the input coefficient bit lengths, intermediate compositions, geometry certification, and the work spent on attempts that eventually return inconclusive. The pre-elimination first jet addresses one of these costs directly: it can certify the relevant rank before some cobordism compositions are formed.

The existing first-jet observer already has a knot-completion multiplier. It reads a radical

one-matching block *after* scalar elimination. Moving the query earlier is a genuine implementation change, but merely renaming the old multiplier would not be one. The new doubled binary matrix works when scalar coefficients are still present; its invariance under their elimination is proved below. A separate rigidity theorem licenses repeated total-rank substitutions under link completions.

2.3 Why repeated graph setup is a separate cost

At a fixed suffix, signed Tait-graph elimination gives a small terminal response. A singular interior must be represented by both its radical coordinates and their coupling to the boundary. Earlier fixed-stage responses already use this principle. Rebuilding the next suffix and eliminating its interior independently can nevertheless repeat most of the work. The reverse-suffix construction retains the elimination while the black cut-face fragments merge. Its admissibility invariant is that an eliminated closed fragment can never be identified again.

This improvement is orthogonal to the relative-complex state count. A response of bounded dimension can be queried exponentially many times. Conversely, many early-exit inputs do not justify precomputing all suffixes. The theorem in Section 5 states setup and query costs separately, and the experiments time only the scope they actually exercise.

2.4 Parameters and representations

Parameter	Meaning and necessary accounting
n	Crossings of the supplied validated diagram.
$b(D)$	Maximal over-runs of its cyclic over/under traversal; not the knot's minimum bridge number.
r, N	Generator count and total explicit relator letters of a verified presentation.
S, s, t, d	Literal word-circuit input size, reachable gates, retained cut gates, and post-cut formal degree.
w, W	Actual scan frontier and actual suffix dart frontier in their respective constructions.
g	Inclusive largest crossing gap between one-object checkpoints, including initial and terminal segments.
M_s, L_s	Relative-complex object count and explicit coefficient length before or after the stated operation.
p	Field characteristic for modular determinant responses; bit cost is separate when p varies.

Several quantities can be measured inexpensively, but an observed small value is not a universal bound. The experiments store actual diagrams and orders so these distinctions can be checked on the same input. Theoretical claims always specify the representation; a binary exponent is not charged as one explicitly expanded relator letter.

3 An exact algebraic route controlled by presentation size

The reduction from knottedness to real algebraic feasibility is established work: Meesum and Prathamesh gave a constant-degree formulation with four real variables per Wirtinger generator [14]. The issue addressed here is how to reduce the number of independent quaternion variables without hiding an exponential increase in polynomial degree or coefficient size. We prove three quantitative compilation results. They concern explicit words, straight-line word circuits, and a hybrid representation that retains selected circuit nodes. None presupposes a fast algorithm for finding an optimally small presentation.

All complexity statements below refer to deterministic bit operations. The notation $A^{O(B)}$ has a universal implicit constant. A polynomial factor in an input length always has a universal exponent

independent of the number of generators. A knot-group presentation is supplied with a verified derivation, or an explicit knot-group promise when stating a conditional theorem. The cost of establishing that origin is discussed separately in Section 3.8.

3.1 The topological and real-algebraic ingredients

The only deep knot-theoretic input needed in this section is the following consequence of Kronheimer–Mrowka.

Theorem 3.1 (Kronheimer–Mrowka consequence). *For a tame knot $K \subset S^3$,*

$$K \text{ is nontrivial} \iff \pi_1(S^3 \setminus K) \text{ admits a homomorphism to } \mathrm{SU}(2) \text{ with nonabelian image.}$$

Proof. The unknot group is infinite cyclic, so every image of it is abelian. For the other direction, apply [8, Theorem 1] to +1 surgery on a nontrivial knot. The resulting fundamental group has a homomorphism to $\mathrm{SU}(2)$ with noncyclic image. Its first integral homology is zero, so an abelian image would be trivial. Consequently the image is nonabelian. Composing with the quotient map from the knot group gives the required representation. This argument explains why the distinction between “noncyclic” and “nonabelian” in the cited theorem causes no difficulty. $\square$

We use the following established form of existential real decision.

Theorem 3.2 (Existential real decision bound). *Let a quantifier-free formula of bounded Boolean length involve a fixed number of integer polynomials in $k \geq 1$ real variables, each of total degree at most $D \geq 2$, with coefficient bit length at most τ . Existence of a real solution can be decided deterministically in*

$$\mathrm{poly}(\tau + k) D^{O(k)}$$

bit operations, with the dense polynomial input included in the bound. Strict inequalities are allowed.

This is a standard consequence of Renegar’s existential decision theorem [22]. An independently accessible account is [3, Theorem 2.18]: for one quantified block and no free variables, the arithmetic-operation bound is $(sD)^{O(k)}$, and the bit length of intermediate integers is at most $\tau D^{O(k)}$. Here s is the number of input polynomials. Using ordinary integer arithmetic and fixing s gives the displayed bit bound. The dense-input convention in [3, Section 2.5] is essential. The proofs below include the cost of producing that dense input; a short arithmetic circuit alone is not treated as a short dense polynomial.

3.2 Unit quaternions and the correct nonabelian test

Write a quaternion as $q = (a, \mathbf{v})$, with $a \in \mathbb{R}$ and $\mathbf{v} \in \mathbb{R}^3$. Its product and conjugate are

$$(a, \mathbf{v})(b, \mathbf{w}) = (ab - \mathbf{v} \cdot \mathbf{w}, a\mathbf{w} + b\mathbf{v} + \mathbf{v} \times \mathbf{w}), \quad (1)$$

$$\overline{(a, \mathbf{v})} = (a, -\mathbf{v}). \quad (2)$$

The unit quaternions, defined by $a^2 + \|\mathbf{v}\|^2 = 1$, form $\mathrm{SU}(2)$. Conjugation gives inversion on this group. In particular, inverse letters require linear polynomials, not division.

Let $x_1, \dots, x_r$ be the original generators of a presentation and let $q_i = (a_i, \mathbf{v}_i)$ be their putative images. Define

$$H(q_1, \dots, q_r) = \sum_{1 \leq i < j \leq r} \|\mathbf{v}_i \times \mathbf{v}_j\|^2. \quad (3)$$

This is an integer polynomial of degree at most four. The empty sum is zero.

Lemma 3.3 (Nonabelian-image test). *A representation of a group generated by $x_1, \dots, x_r$ has nonabelian image if and only if $H > 0$.*

Proof. Equation (1) gives $q_i q_j - q_j q_i = (0, 2\mathbf{v}_i \times \mathbf{v}_j)$. Thus q_i and q_j commute exactly when their cross product vanishes. A group generated by a pairwise commuting set is abelian. Finally, a sum of real squares is positive exactly when at least one of its summands is nonzero. $\square$

Lemma 3.4 (Conjugate meridional generators). *Suppose the generators $x_1, \dots, x_r$ are pairwise conjugate in the presented group. For every representation their scalar quaternion parts are equal, and the lower-degree polynomial*

$$H_{\text{mer}} = \sum_{i=2}^r \|q_i - q_1\|^2$$

is positive exactly when the image is nonabelian.

Proof. Quaternion conjugation preserves the scalar part, so the asserted scalar equalities hold. If the entire image is abelian, the images of the words conjugating x_1 to the other generators commute with q_1 ; consequently all generator images equal q_1 . Conversely, if all generator images are equal, they generate a cyclic, hence abelian, subgroup. The sum-of-squares test expresses the negation of these equalities. $\square$

The hypothesis concerns conjugacy *in the presented group*, not just the fact that selected quaternion values happen to have the same trace. For a complete Wirtinger presentation of a knot, or a certified generating set of consistently oriented meridians obtained from it, it applies. That shortcut does not survive arbitrary changes of generators. For example,

$$\langle a, b \mid b = a^2 \rangle \cong \mathbb{Z}$$

is an unknot-group presentation. Sending a to the quaternion i and b to -1 gives distinct generator images and still gives an abelian image. The test (3) remains correct for this presentation. Auxiliary variables introduced during compilation are word values, not additional independent generators; it suffices to test the original r generators.

Lemma 3.5 (A universal simultaneous-conjugation gauge). *Every tuple $q_1, \dots, q_r$ of unit quaternions with $r \geq 2$ is simultaneously conjugate to a tuple of the form*

$$q_1 = (a_1, b_1, 0, 0), \quad q_2 = (a_2, b_2, c_2, 0),$$

with the later quaternions unrestricted. The assertion includes central q_1 , central q_2 , and parallel imaginary parts. For $r = 1$, the first displayed condition alone is always attainable. Hence imposing these coordinate zeros loses no solution to the representation and nonabelian-image conditions above.

Proof. Let $\mathbf{v}_1, \mathbf{v}_2$ be the first two imaginary parts. If $\mathbf{v}_1 \neq 0$, choose the first vector of a positively oriented orthonormal frame to be $\mathbf{v}_1 / \|\mathbf{v}_1\|$. If the projection of $\mathbf{v}_2$ to its orthogonal plane is nonzero, choose the second frame vector in that projection's direction; otherwise choose any perpendicular unit vector. The third vector is the cross product of the first two. Coordinates in this frame put $\mathbf{v}_1$ on the first axis and $\mathbf{v}_2$ in the first two coordinate directions. If $\mathbf{v}_1 = 0$, choose the first frame vector along $\mathbf{v}_2$ when it is nonzero, and choose any positively oriented orthonormal frame when both vectors vanish. This also has the required properties. The case $r = 1$ uses just the first-vector construction.

The resulting coordinate change is in $\text{SO}(3)$. Every such rotation is induced by conjugation by a unit quaternion: rotation through angle θ about the unit vector $\mathbf{n}$ is realized by $u = (\cos(\theta/2), \mathbf{n} \sin(\theta/2))$, as follows by substituting in (1). Replace every generator and every auxiliary word value by uqu^{-1} . Conjugation preserves unit norms, word equations, and commutation or equality of generator images. The scalar parts are unchanged. Thus all constraints are preserved. $\square$

No coordinate is required to be strictly positive or nonzero in this gauge. It removes three real generator coordinates when $r \geq 2$, and two when $r = 1$. If the conjugate-meridian hypothesis also holds, one may additionally share a single scalar variable. For $r \geq 2$ the generator tuple then uses $3r - 2$ real variables; the auxiliary word variables still use four each. The general bounds below use the simpler upper bound of four variables per quaternion, so these safe reductions can only improve them.

Theorem 3.6 (Known traceless-meridian detection). *A knot $K \subset S^3$ is nontrivial if and only if its group has a nonabelian $\text{SU}(2)$ representation carrying a chosen meridian to the quaternion $i = (0, 1, 0, 0)$.*

The forward implication is the established theorem of Kronheimer–Mrowka, [9, Corollary 7.17]; the reverse implication follows because the unknot group is cyclic. “Irreducible” in that source is equivalent to nonabelian image for representations into $\text{SU}(2)$. This is a stronger published ingredient than the unrestricted detection consequence used to prove the general presentation bounds.

Corollary 3.7 (A smaller formula for certified meridians). *For a knot-group presentation on $r \geq 2$ certified consistently oriented meridians, exact nontriviality can be tested while imposing*

$$q_1 = (0, 1, 0, 0), \quad q_2 = (0, b_2, c_2, 0), \quad q_j = (0, b_j, c_j, d_j) \quad (j \geq 3).$$

With u retained auxiliary quaternions this uses $3r - 4 + 4u$ real variables. The degree and coefficient estimates of the subsequent compilation theorems do not increase.

Proof. Theorem 3.6 fixes the first meridian image to i . Every other meridian is conjugate to it in the knot group, so all scalar parts vanish. A remaining rotation about the first imaginary axis puts the second imaginary vector in the first two coordinate directions, including the parallel case. The second quaternion contributes two variables, the later $r - 2$ each contribute three, and each auxiliary contributes four. Substitution of zero and one cannot increase total degree or coefficient ℓ^1 norm, so the coefficient-height estimates proved below remain valid. $\square$

This corollary is not a new trace-existence theorem. It describes a sound further specialization of the meridional compiler. The recorded implementation retains its general gauge and uses trace zero only as a SAT-witness probe; it does not implement this smaller complete formula. Consequently the reported dimensions and timings below are not those of the stronger specialization. For arbitrary generators, a chosen generator need not represent a meridian, and neither the trace restriction nor the fixed value i is justified without that additional provenance.

3.3 Explicit relators: a degree–variable tradeoff

Let

$$\mathcal{P} = \langle x_1, \dots, x_r \mid w_1, \dots, w_m \rangle, \quad r \geq 1,$$

be a finite presentation. The relators are explicit words in the generators and their inverses, and

$$N = \sum_{j=1}^m |w_j|$$

is their total letter length. Empty relators are discarded. This standard normalization ensures $m \leq N$ if $N > 0$; otherwise there are no relators. Encoding a generator index costs $O(\log(r+1))$ bits, so the input length is polynomial in $N+r$. If an input format permits an arbitrarily long list of empty relators, its literal parsing time must be added separately.

Theorem 3.8 (Explicit-presentation tradeoff). *For any integer $L \geq 1$, existence of a nonabelian $SU(2)$ representation of $\mathcal{P}$ can be decided in*

$$\text{poly}(N+r) (L+2)^{O(r+N/L)} \quad (4)$$

bit operations, where one may restrict $L \leq \max(1, N)$. In particular, it can be decided in

$$\text{poly}(N+r) \left(2 + \frac{N}{r}\right)^{O(r)}. \quad (5)$$

If $\mathcal{P}$ presents the group of a supplied knot K , these are also bounds for recognizing whether K is the unknot, with the presentation-provenance cost added.

Proof. Partition each nonempty relator w_j into consecutive chunks

$$w_j = c_{j,1} \cdots c_{j,h_j}, \quad h_j = \lceil |w_j|/L \rceil, \quad 1 \leq |c_{j,a}| \leq L.$$

Introduce a quaternion variable $y_{j,a}$ for each intermediate prefix $c_{j,1} \cdots c_{j,a}$, with $1 \leq a < h_j$. No such variable is needed for the empty prefix or the final prefix, whose values are prescribed to be 1. The number u of auxiliary quaternion variables satisfies

$$u = \sum_j (h_j - 1) \leq N/L.$$

There are consequently $k = 4(r+u)$ real variables.

Let $C_{j,a}$ denote the quaternion polynomial obtained by evaluating the letters of $c_{j,a}$ using the original variables q_i and their conjugates. Each coordinate of $C_{j,a}$ has degree at most L . Impose the quaternion equations

$$y_{j,1} = C_{j,1}, \quad y_{j,a} = y_{j,a-1} C_{j,a} \quad (1 < a < h_j), \quad 1 = y_{j,h_j-1} C_{j,h_j}.$$

When $h_j = 1$, use just $1 = C_{j,1}$. Each quaternion equation means four real equations, of degree at most $L+1$. Also impose the unit-norm equation on every original and auxiliary quaternion. The latter equations are redundant for correctly evaluated auxiliary variables, but their inclusion is harmless and makes the intended domain explicit.

For any assignment of unit quaternions to the original generators, the prefix equations have a unique assignment to all the auxiliary variables. This assignment satisfies the final equation for w_j if and only if $w_j(q_1, \dots, q_r) = 1$. This proves that the equations encode precisely the representations of the original presented group.

List the scalar polynomial residuals of all equations as $p_1, \dots, p_A$ and set

$$F = \sum_{\nu=1}^A p_\nu^2.$$

Over the real numbers, $F = 0$ is equivalent to all the original equalities. The required feasibility formula therefore has only two polynomial atoms:

$$\exists(q, y) \quad F(q, y) = 0 \wedge H(q) > 0. \quad (6)$$

Its maximum degree is at most $D = 2L + 2$, which is at least four.

We now account for bit size and dense compilation. A coordinate of a product of ℓ quaternion letters has coefficient ℓ^1 norm at most $4^{\ell-1}$ for $\ell \geq 1$: at each multiplication, one coordinate is a signed sum of four products. Conjugation only changes signs. An auxiliary prefix factor increases the word length by at most one. Thus each residual has coefficient ℓ^1 norm $2^{O(L)}$, and squaring it preserves this form of bound. There are $O(N+r)$ residuals. Every coefficient of F or H consequently has bit length

$$\tau = O(L + \log(N + r + 2)).$$

This argument allows coefficients to accumulate when identical monomials are collected; it does not assume that aggregate coefficients remain in a fixed finite set.

A polynomial of degree at most D in k variables has at most

$$\binom{k+D}{k} \leq (D+1)^k$$

monomials. Expand each chunk by successive multiplication by linear quaternion letters, multiply by its optional prefix variable, square the residuals, and collect equal monomials using exact integer arithmetic. Even naive dense multiplication costs at most a polynomial in the number of monomials. All dense polynomials can therefore be constructed in

$$\text{poly}(N + r + L) (D + 1)^{O(k)}$$

bit operations. For $L \leq \max(1, N)$ this has the form (4). Applying Theorem 3.2 to (6) has the same bound. This includes both compilation and decision.

Finally take $L = \max(1, \lceil N/r \rceil)$. Then $u \leq r$ when $N \geq r$, while $u \leq N \leq r$ when $N < r$, and $L + 2 = O(2 + N/r)$. This gives (5). When $N = 0$ there are only the original quaternion variables and the same argument with $L = 1$ applies. The knot corollary follows from Theorem 3.1. $\square$

Remark 3.9 (What the tradeoff means). The two extreme formulations are $L = 1$, with constant degree and $O(N + r)$ quaternion variables, and $L = N$, with no prefix variables and degree $O(N)$. Introducing some prefix variables can improve the asymptotic bound over complete substitution. The optimized choice is a guarantee, not a recommendation to use that value in every implementation: the actual support of the polynomials and the individual relator lengths may justify a different value.

3.4 A concrete consequence for the supplied diagram

For a nonempty oriented knot diagram D , traverse its unique component and record an O at an overcrossing encounter and a U at an undercrossing encounter. Define $b(D)$ to be the number of maximal O -runs in this cyclic word. This is the number of maximal diagrammatic overpasses, not the minimum bridge number of the knot type.

Lemma 3.10 (Overpass presentation). *From an n -crossing knot diagram with $b = b(D)$, one can construct a presentation of its knot group with b meridional generators and total explicit relator letter length at most $2n + 2b$. The construction and a check of the elimination steps take polynomial time in n .*

Proof. Choose a meridian x_i for each maximal overpass. Along the intervening underpass, each crossing conjugates the current meridian by the meridian of the overpassing strand, with the sign determined by the oriented crossing convention. If the successive conjugating letters are $z_1, \dots, z_\ell$, all of them are among the selected overpass generators and their inverses. Eliminating the intermediate Wirtinger generators gives one relation

$$x_{\text{next}} = wx_{\text{previous}}w^{-1}, \quad |w| = \ell,$$

with the appropriate order of factors. This elimination uses only relations that define a new intermediate generator from preceding data, so it is a sequence of valid Tietze eliminations. The underpasses partition the n undercrossing encounters, hence $\sum \ell = n$. Writing each of the b resulting relations as a relator gives total length $\sum (2\ell + 2) = 2n + 2b$. $\square$

Corollary 3.11 (Diagrammatic-overpass bound). *Unknot recognition on a supplied n -crossing diagram D can be performed deterministically in*

$$\text{poly}(n) \left(2 + \frac{n}{b(D)} \right)^{O(b(D))}.$$

In particular, the bound is polynomial for bounded $b(D)$ and is $n^{O(\log n)}$ when $b(D) = O(\log n)$. For arbitrary diagrams it is at most $2^{O(n)}$ times a polynomial factor.

Proof. Apply Theorem 3.8 and Lemma 3.10. The last assertion follows from $b \log(2 + n/b) = O(n)$ for $1 \leq b \leq n$. $\square$

This result is deliberately parameterized by the supplied diagram. An alternating diagram has cyclic encounter word $OUOU \cdots OU$, so $b(D) = n$, even when its knot type has bridge number two. The usual alternating diagram of $T(2, m)$ is an example. The corollary does not provide a polynomial-time conversion to a bridge-minimizing diagram. It also does not replace $b(D)$ by the number of strands of a supplied braid. Iterated Artin substitutions can produce very long expanded words; the next theorem is the appropriate bound when these substitutions are kept as circuits.

Remark 3.12 (The optimized bridge compiler). The implementation need not write both copies of every conjugating word. It may instead impose $q_{\text{next}}q_w = q_wq_{\text{previous}}$. Unit generator norms and the prefix definitions ensure that q_w is invertible, so this is precisely the same relation. Split the n conjugator letters into chunks of at most L . There are $u \leq n/L$ prefix quaternion variables. The final conjugacy equation has degree at most $L + 2$: one prefix factor, at most L final letters, and one endpoint generator. Thus the aggregate degree is at most $2L + 4$, while coefficient bit lengths remain $O(L + \log(n + b + 2))$. Lemmas 3.4 and 3.5 reduce the number of real variables to $3b - 2 + 4u$ for $b \geq 2$. When $b = 1$, the nonabelian test is identically zero and the decision is immediate. This more economical compiler retains the asymptotic bound of Corollary 3.11.

3.5 Compressed words: retaining every circuit gate

A straight-line word circuit is a finite directed acyclic graph whose inputs are $x_1, \dots, x_r$ and, optionally, the identity, and whose internal gates are concatenation and inversion. A list of designated outputs specifies the relators. The circuit may share subwords. Let s be the number of internal gates after expanding any binary integer-power instruction into repeated squaring and multiplication. Negative powers also use inversion. Thus a power w^e costs $O(\log(|e| + 1))$ gates; it is not counted as one constant-degree gate regardless of e . Let S be the bit length of the whole circuit input, including all output references and exponent encodings.

Theorem 3.13 (Straight-line presentation bound). *For a presentation specified by a straight-line word circuit with r generators and s elementary internal gates, existence of a nonabelian $SU(2)$ representation can be decided in*

$$\text{poly}(S + r + s) 2^{O(r+s)}$$

bit operations. The feasibility formula has $4(r + s)$ real variables, one equality and one strict inequality, and maximum degree four. Its aggregate coefficient bit length is $O(\log(S + r + s + 2))$.

Proof. Assign a quaternion variable to every generator and every internal gate. For a concatenation gate $z = uv$, impose the four coordinate equations $q_z - q_u q_v = 0$. They have degree at most two. For an inversion gate $z = u^{-1}$, impose $q_z - \overline{q_u} = 0$, which is linear. Identity inputs are constants. Impose $q_z = 1$ for each relator output and unit-norm equations for all quaternion variables. Induction in a topological ordering of the circuit proves that every gate variable has the unique correct value for the chosen generator images. The equations therefore encode exactly the original group representations, regardless of how long the expanded words would be.

Square and sum all scalar residuals to form F , and retain $H > 0$ from (3). Both F and H have degree at most four. Every residual has a bounded number of monomials and bounded integer coefficients, even when a gate repeats the same input. Collecting terms from $O(S + r + s)$ residuals, and from $O(r^2)$ terms in H , gives coefficient bit length $O(\log(S + r + s + 2))$. For fixed degree four, the number of monomials in $4(r + s)$ variables is polynomial in $r + s$, so even full dense compilation is polynomial in the input parameters. Theorem 3.2 then gives the bound. $\square$

Remark 3.14 (A circuit gate is not a free degree reduction). The chain $z_0 = x$, $z_{j+1} = z_j z_j$ computes x^{2^s} with s gates. Eliminating every gate can leave coordinate polynomials of formal degree 2^s . Retaining the gates gives quadratic local constraints but adds four real variables per gate. Theorem 3.13 pays for those variables. It does not infer quasipolynomial time from a small number of original generators when the circuit has linear size in the crossing count.

3.6 A hybrid bound for a cut in the word circuit

Full expansion and full gate retention are special cases of one construction. Choose t internal gates to retain as quaternion variables. For every retained gate, expand its defining word through unretained gates until another retained gate or an original generator is reached. Do the same for every relator output. In this expansion, inversion reverses the word and inverts its letters. Identity letters are discarded. Let $d \geq 1$ bound the length of every resulting word in the cut variables and their inverses. Equivalently, it is enough to bound the formal degree obtained by assigning degree one to generator and cut variables, degree zero to identity, taking sums at concatenation gates, and preserving degree at inversion gates. For a retained gate's own definition, apply its operation before stopping at upstream cut gates.

Theorem 3.15 (Hybrid cut bound). *With the notation above, a supplied choice of t retained gates and degree bound d gives a deterministic decision algorithm with bit complexity*

$$\text{poly}(S + r + s) (d + 2)^{O(r+t)}.$$

The cost of expanding and collecting all relevant coordinate polynomials is included. The theorem is for word circuits with concatenation, inversion, and identity constants; it does not assert the same coefficient-height bound for unrestricted arithmetic circuits.

Proof. Use quaternion variables only for the original generators and the retained gates. Each retained gate is constrained to equal the polynomial word in upstream cut variables given by its defining expansion. Each relator word is constrained to be the identity. Add unit-norm equations and the nonabelian test on the original generators. Topological induction over retained gates proves equivalence with the full circuit equations: there are no circular definitions, and unretained gates are uniquely evaluated from their inputs.

After summing squares, there are two polynomial atoms in $k = 4(r + t)$ variables, with maximum degree at most $\max(4, 2d)$. Every expanded expression is a genuine word of length at most d in quaternion variables or conjugates. Its coefficient ℓ^1 norm is at most 4^{d-1} , so the aggregate coefficient bit length is $O(d + \log(S + r + s + 2))$. In particular, repeated squaring of arbitrary integer constants cannot create an unaccounted height explosion: such constants and operations are not part of the word-circuit language.

Memoize the polynomial of each relevant unretained gate, treating each retained gate as a new variable when used by a later gate. Store its defining polynomial separately. All intermediate nonconstant formal degrees are at most d , since concatenation adds nonnegative degrees and inversion preserves them. A dense polynomial of degree $2d + 4$ in k variables has at most $(2d + 5)^k$ monomials. Ordinary exact dense multiplication, summation, and coefficient collection therefore cost

$$\text{poly}(S + r + s + d)(d + 2)^{O(k)}.$$

The same bound covers existential real decision. Since $r \geq 1$, every fixed polynomial factor in d is absorbed into $(d + 2)^{O(r+t)}$. This gives the claimed form, including dense compilation. $\square$

A useful, fully computable size objective for a proposed cut is

$$\Lambda = (r + t) \log_2(d + 2).$$

It reflects the exponent in the proved worst-case bound, while practical decisions can additionally use monomial counts and coefficient heights. Optimizing Λ over all cuts is a separate combinatorial problem; the theorem assumes the cut is supplied. One can always evaluate a fixed list of cuts and keep the best proved estimate, or run their solvers in a fair portfolio. No exhaustive search over cuts is required for correctness.

Remark 3.16 (A greedy threshold gives a certified cut). A simple implementable rule processes a concatenation DAG in topological order. Fix a threshold $h \geq 1$. Store degree at most h for every currently expanded value. At a product of two stored values, compute the sum of their formal degrees. If it exceeds h , retain the product gate, record its defining equation, and reset its stored degree to one; otherwise leave it expanded. Every recorded gate definition has degree at most $2h$, and every output value has degree at most h . Consequently the rule supplies a valid cut with $d \leq 2h$ in Theorem 3.15, and its aggregate formula degree is at most $4h$. The threshold is a bound on stored expressions, not on a retained gate's defining polynomial. For $h = 1$, all nontrivial products are retained and the aggregate degree is at most four. This argument also accounts for the temporary multiplication performed before deciding to retain a gate; that multiplication never exceeds degree $2h$.

Corollary 3.17 (A sufficient quasipolynomial compilation condition). *Suppose an n -crossing knot diagram can be converted, with verified group-preserving provenance, to a word circuit and a cut in $\text{poly}(n)$ time. Suppose the circuit description has polynomial length and*

$$(r + t) \log(d + 2) = O((\log n)^2).$$

Then this conversion followed by exact existential real decision recognizes the unknot in $n^{O(\log n)}$ time. In particular, it is enough to have $r + s = O((\log n)^2)$ with every gate retained.

This is a sufficient condition on an explicitly constructed representation. No assertion that every knot diagram admits such a construction is proved here. In fact, requiring a small presentation of the *entire* knot group is too restrictive for a universal route, as the following elementary obstruction shows.

Proposition 3.18 (A rank obstruction to universal presentation compression). *For every $m \geq 1$, there is a knot with a $3m$ -crossing diagram whose group has minimum generator number exactly $m + 1$. Consequently no group-preserving compiler can satisfy $(r + t) \log(d + 2) = O((\log n)^2)$ on every n -crossing diagram solely by choosing a presentation of its entire knot group and a cut in its relator circuit.*

Proof. Let K_m be the connected sum of m trefoils, drawn by joining m three-crossing trefoil diagrams without introducing new crossings. The Wirtinger presentation of a trefoil is $\langle a, b \mid aba = bab \rangle$, with a, b meridians. Identifying the meridian a in the connected sum, either by the Wirtinger construction or van Kampen's theorem, gives

$$G_m = \langle a, b_1, \dots, b_m \mid ab_i a = b_i a b_i \ (1 \leq i \leq m) \rangle.$$

Let $V = \mathbb{F}_3^m$ with standard basis $e_1, \dots, e_m$, and let $\text{Dih}(V) = V \rtimes C_2$ with multiplication

$$(v, \epsilon)(w, \delta) = (v + (-1)^\epsilon w, \epsilon + \delta \bmod 2).$$

The assignment $a \mapsto (0, 1)$ and $b_i \mapsto (e_i, 1)$ respects the relators: the images of $ab_i a$ and $b_i a b_i$ are respectively $(-e_i, 1)$ and $(2e_i, 1)$, and these coincide over $\mathbb{F}_3$. It is surjective because ab_i maps to $(-e_i, 0)$, and these elements generate V .

The group $\text{Dih}(V)$ requires $m + 1$ generators. To see the lower bound, start with any generating set of size r and choose one generator z outside V . Such an element exists and has order two. Replace each other generator outside V by its product with z ; retain the generators already in V . These replacements preserve the generated subgroup and leave z together with at most $r - 1$ vectors in V . Conjugation by z negates every vector of V . Therefore the rotation subgroup generated by this set is just the $\mathbb{F}_3$ span of those at most $r - 1$ vectors. To obtain all of V requires $r - 1 \geq m$. The displayed quotient then gives $d(G_m) \geq m + 1$, and the original presentation gives the reverse inequality. Since $t \geq 0$ and $d \geq 1$, the proposed cut objective is then $\Omega(m) = \Omega(n)$ on this family, regardless of relator compression. $\square$

This obstruction concerns the requirement to preserve the whole group, not the complexity of deciding the unknot. Connected sums admit a decision decomposition: a connected sum is trivial exactly when all its summands are trivial. A universal strategy using the algebraic route would therefore need to exploit such certified decompositions, or other logically justified reductions, before demanding small presentation parameters. Finding and certifying a decomposition also has a cost.

Proposition 3.19 (Prime knots also obstruct universal rank compression). *For every odd $m \geq 3$, the pretzel knot $P_m = P(3, \dots, 3)$ with m entries has a prime alternating $3m$ -crossing diagram, and its knot group requires at least m generators.*

Proof. Use the standard pretzel diagram consisting of a cyclic row of m vertical three-half-twist boxes, with adjacent upper endpoints joined above the boxes and adjacent lower endpoints joined

below them. Each box exchanges its two strands. Traversing through two successive boxes advances the upper connecting-arc index by two modulo m . Since m is odd, this visits every upper connecting arc and the diagram has one component. It has $3m$ crossings and is alternating. One of its checkerboard graphs consists of m internally disjoint paths of length three between the same two vertices. This graph has no loop and no cut vertex. Hence its medial diagram is prime: a circle meeting the medial diagram twice and having crossings on both sides would give either a loop or a separating vertex of the checkerboard graph. The theorem that a prime alternating diagram represents a prime link, due to Menasco [15], now gives primeness; see also the extension proved in [21, Theorem 10.3], whose hypotheses specialize to a prime alternating diagram on $S^2 \subset S^3$.

We construct a quotient of the knot group. Fox colorings with values in a vector space V over $\mathbb{F}_3$ satisfy $v_{\text{out}} = 2v_{\text{over}} - v_{\text{in}}$ at each undercrossing. With one consistent choice of two-strand crossing convention, the transformation of a pair of colors through a half twist is

$$T = \begin{pmatrix} 2 & -1 \\ 1 & 0 \end{pmatrix} = I + N, \quad N^2 = 0.$$

Thus $T^3 = I$ over $\mathbb{F}_3$; reversing the half-twist convention replaces T by T^{-1} and has the same consequence. Every twist box therefore preserves its pair of upper colors at the corresponding lower positions. The exterior upper and lower connections have the same cyclic pattern. Consequently the m upper connecting arcs can receive arbitrary values, and these values extend to all arcs of the diagram without further constraints.

Take $V = \mathbb{F}_3^{m-1}$ and assign the m upper connecting arcs the values $0, e_1, \dots, e_{m-1}$. Map each Wirtinger meridian with color v to the reflection $(v, 1)$ of $\text{Dih}(V)$. At a crossing,

$$(w, 1)(v, 1)(w, 1) = (2w - v, 1),$$

so every Wirtinger relation holds; the sign of the conjugating meridian does not matter since its image is an involution. The image contains $(0, 1)$ and $(e_i, 1)$ for every i , hence contains all rotations and is surjective. The generator-rank argument in the preceding proof, with $\dim V = m - 1$, implies that this quotient, and therefore the knot group, requires at least m generators. $\square$

Neither proposition is a lower bound on unknot-recognition time. In particular, these families have nonconstant Fox colorings that already certify knottedness by elementary linear algebra. The prime example shows that prime decomposition alone cannot make whole-group generator rank uniformly small. The useful research target is instead a portfolio of certified decision reductions, followed by controlled presentation or circuit-cut parameters on the instances that remain. The real-algebraic implication for any such residual instance is proved above; establishing a universal bound for a residual class requires a separate theorem about that class. The rank arguments are included as explicit obstructions for this compilation strategy; no priority claim is made for the knot-group rank bounds or the use of dihedral quotients. Lustig–Moriah’s more general ordinary-rank calculations for generalized Montesinos knots [13] are recalled in the author’s account [16, Theorem 3.6]. The elementary proof above makes the specific obstruction needed here independent of that more general theory.

3.7 Cyclic images, gauge choices, and exact solver outcomes

Several implementation restrictions follow directly from the proofs.

First, the unit equations are indispensable for interpreting quaternion conjugation as inversion. In contrast, an additional norm equation at a gate is logically redundant if all its ancestors are

correctly constrained and the original generators are unit quaternions. Omitting those redundant equations is safe only with that inductive invariant preserved.

Second, gauge fixing means simultaneous conjugation. It can rotate imaginary quaternion axes, but it preserves each generator’s scalar part and hence its trace. Setting a meridian equal to the identity is not a gauge choice and destroys all nonabelian representations of a Wirtinger knot group. Setting its scalar part to zero is also not a gauge choice, but Theorem 3.6 makes it a complete restriction for verified knot meridians. Fixing that meridian to i is then legitimate as in Corollary 3.7. The present generic solver interface uses only a trace-zero witness probe: a satisfying model is decisive, and failure returns to the unrestricted query. Common scalar parts may be imposed on a certified consistently oriented meridional generating set, since those generators are conjugate, but not on arbitrary generators or prefix gates.

Third, the decision formula is existential over the reals. A satisfying assignment certifies a nonabelian representation and therefore knottedness. Infeasibility certifies the unknot only for a correctly constructed formula with verified knot-group provenance. A solver timeout, resource limit, unknown result, or failure to find a numerical approximation is not an infeasibility result. Approximate solutions alone are not exact witnesses.

Finally, Theorems 3.8, 3.13, and 3.15 use a theoretical singly exponential existential real decision procedure. A practical implementation may instead pass the same exact integer-polynomial formula to NLSAT, cylindrical algebraic decomposition, or another sound exact backend. This transfers the logical reduction, but does not transfer Renegar’s worst-case running-time bound to that implementation. A benchmark of the practical backend and the asymptotic existence theorem answer different questions and must be reported separately.

3.8 Presentation provenance and novelty boundaries

For a diagrammatic-overpass presentation, provenance is the explicit Wirtinger elimination in Lemma 3.10. A more ambitious presentation simplifier must supply comparably checkable evidence that its output presents the same knot group. A sequence of Tietze transformations is one possible form. Each retained or eliminated generator must have its defining relation accounted for, and every change to a relator must be justified. The verifier must retain the correct relators and the correct word multiplication order. A homology calculation, equality of a few knot invariants, or successful tests on a finite knot table does not establish this group isomorphism.

If provenance costs T_{prov} bit operations to construct or verify, add T_{prov} to the applicable bound. A short presentation that exists without a bounded-cost construction does not give an algorithm with the same bound. Likewise, a small number of generators does not help if explicitly writing its relators has already cost exponential time and no smaller circuit is supplied.

The underlying $\text{SU}(2)$ reduction, the sum-of-squares aggregation, and singly exponential real decision are established ingredients. The 2018 paper explicitly suggests substitution to reduce variable count as a possible direction for further work. The present contribution is the fully costed presentation-sensitive tradeoff, its concrete consequence for the overpasses of the supplied diagram, and the circuit/cut formulations that preserve word compression without concealing degree. A targeted literature search did not locate the exact bound $\text{poly}(N + r)(2 + N/r)^{O(r)}$ for knot presentations, but absence from that search is not a priority proof. These results should be presented as proved refinements of the established reduction, not as a claim that the general unknot-recognition problem has been solved in a new complexity class.

3.9 Source corrections relevant to implementation

The arXiv v1 text of [14] contains several local errors that do not invalidate its intended reduction but should not be copied into code. Its preliminary definition calls the matrices in $SU(2)$ Hermitian; the intended word is unitary. Its displayed quaternion-shaped matrix parameterization and unit-norm condition use the correct group. The declared convention for Algorithm 1 is that “Yes” means unknot, but its satisfiable branch returns “Yes” for a formula that detects knottedness. Under that declared convention, the two output labels must be interchanged. Finally, after all squared residuals are added and like terms are collected, integer coefficients need not stay in the fixed small set claimed in its complexity discussion. A logarithmic coefficient-height bound in the number of residuals suffices for its $2^{O(n)}$ conclusion and is the bound used above.

4 First-jet decisions before cobordism elimination

The maintained closure observer reads a radical complex after all scalar units have been cancelled. It deliberately declines a block containing a scalar coefficient. This section moves the observation before that operation and proves a rigidity statement that removes the knot-completion hypothesis when the multiplier is one. The result is an optional decision scanner; no original graded homology is claimed after a replacement.

Fix a nonempty matching m with p arcs and put

$$R = \mathbb{F}_2[x_1, \dots, x_p]/(x_1^2, \dots, x_p^2), \quad A_\epsilon = \mathbb{F}_2[\epsilon]/(\epsilon^2).$$

Let $C = (R \otimes V, Q)$ be a bounded finite free complex supported entirely on m . It must be a whole relative direct summand: no omitted differential entry may attach it to another object. Write

$$Q = A + \sum_{i=1}^p x_i B_i + \text{terms of dot degree at least two}, \quad B = \sum_i B_i, \quad N = \dim V.$$

All matrices raise homological degree by one. The identities $A^2 = 0$ and $AB + BA = 0$ follow from $Q^2 = 0$. They are necessary first-jet tests, not sufficient tests of the whole polynomial identity $Q^2 = 0$.

4.1 Two ordinary binary ranks

Let $r(M)$ denote the total rank of a matrix on the direct sum of all homological degrees, equivalently the sum of its adjacent-degree ranks. Define

$$t = N - 2r(A), \quad \kappa = N - r \begin{pmatrix} A & 0 \\ B & A \end{pmatrix}. \quad (7)$$

Proposition 4.1 (Pre-elimination first-jet invariants). *The number t is the number of objects in a completely scalar-reduced one-matching complex. If B_* is induced by B on $H(V, A)$, then*

$$\kappa = t - r(B_*).$$

Both t and κ are unchanged by scalar-unit cancellation. Moreover $\kappa = 0$ if and only if $t = 0$. If $t > 0$, then $1 \leq \kappa \leq t$.

Proof. The homomorphism $R \rightarrow \mathbb{F}_2$ setting every x_i to zero sends every scalar-unit cancellation to a binary cancellation. At termination its differential is zero, so its object count is $\dim H(V, A) = t$. This is the existing residue-survivor invariant specialized to one matching.

The homomorphism $R \rightarrow A_\epsilon$ sending every x_i to ϵ sends Q to $A + \epsilon B$. In the basis $1, \epsilon$ its underlying binary differential is the doubled matrix in (7); its homology dimension is 2κ . Scalar-unit cancellation remains a homotopy equivalence after substitution, so κ is invariant without any cobordism-valued Schur update.

Choose a binary special contraction of (V, A) onto $H(V, A)$. Transferring ϵB retains only its first perturbation term because $\epsilon^2 = 0$. The transferred differential is ϵB_* and its binary rank is $r(B_*)$. Its homology dimension is $2t - 2r(B_*)$, giving the asserted formula.

Regard B_* as a representation of a finite linearly oriented quiver; it need not square to zero. The number of intervals in its decomposition is $t - \sum_h r(B_{*,h}) = \kappa$. A nonzero bounded quiver has at least one interval. If $t = 0$, scalar-unit cancellation removes the whole relative complex, so it is contractible. $\square$

Lemma 4.2 (Established knot-completion multiplier). *If the actual classical completion L_m is a knot, then*

$$\dim H(\text{cl}(C)) = \kappa \dim \text{Kh}(L_m; \mathbb{F}_2). \quad (8)$$

This is the closure formula already established in the repository's report 28 and `synthesis/closure_resets.tex`. We reproduce its chain-level justification to make the pre-elimination query self-contained [19]. The new all-link extension at $\kappa = 1$ is proved separately below.

Proof. Let (D, d_D) be the ordinary Khovanov complex of L_m . Its marked-arc dot maps X_i commute with each other and with d_D , and satisfy $X_i^2 = 0$. On $D^a \otimes V^b$ use total cohomological degree $a + b$. The completed differential is $\mathcal{D} = d_D \otimes 1 + Q(X_1, \dots, X_p)$; both summands have total degree one. All tensor signs disappear over $\mathbb{F}_2$.

Consider moving the mark for variable x_i across one crossing, from edge u to the continuation edge v of the same link strand. The local reverse-saddle map H has cohomological degree -1 , satisfies $H^2 = 0$, commutes with every fixed dot map, and obeys

$$d_D H + H d_D = X_u + X_v = \delta, \quad H \delta = \delta H = 0.$$

These are strict chain identities, not merely identities on homology: Nahm records them in Lemma 2.4, with attribution to Hedden–Ni, Baldwin–Levine–Sarkar, and Lipshitz–Sarkar [18]. The last identity also follows there by combining $H X_u = X_v H$ with commutation of H with X_u .

Put $P = \partial Q / \partial x_i$ and evaluate its other variables at the current marked edges. It has degree one on V and is independent of x_i . Differentiating $Q^2 = 0$ gives $PQ + QP = 0$. The map HP on $D \otimes V$ has total degree zero, and $G = 1 + HP$ is self-inverse since $H^2 = 0$. Expansion, with compositions acting right to left, gives

$$GDG = \mathcal{D} + \delta P + H d_D H P^2 = \mathcal{D} + \delta P.$$

Here the linear Q -terms cancel by $PQ + QP = 0$, the quadratic Q -terms contain H^2 , and $H d_D H = H \delta = 0$ kills the remaining quadratic term. In particular no assumption $P^2 = 0$ is being made. Since each coefficient is square-free in x_i , adding δP replaces exactly the action X_u by X_v in that variable. If quantum gradings are retained, H has quantum degree -2 and differentiation with respect to x_i raises quantum degree by two, so the same G has quantum degree zero for homogeneous Q .

Because L_m is a knot, every mark can be moved along its strand to one chosen edge by finitely many such conjugations. Their composition is a chain isomorphism, providing coherent transport of the entire polynomial matrix. Once all actions equal X , every monomial of degree at least two vanishes and the differential is $d_D + A + XB$. An equality of dot maps on homology alone would not justify this step.

For completeness, let ν be the operator changing one circle label x to 1 and summing over circles. Over $\mathbb{F}_2$,

$$\nu^2 = 0, \quad \nu d_D = d_D \nu, \quad \nu X + X \nu = 1;$$

see Shumakovitch, Lemma 3.1.C and the proof of Theorem 3.2.A [24]. Thus $E = \text{im } \nu = \ker \nu$ is a subcomplex and $D = E \oplus XE \cong A_\epsilon \otimes E$, with X acting as multiplication by ϵ . The transported completed complex is therefore the tensor product over $\mathbb{F}_2$ of E and $(A_\epsilon \otimes V, A + \epsilon B)$. The latter has homology dimension 2κ by Proposition 4.1; the former has homology dimension $\frac{1}{2} \dim H(D)$. The field Künneth formula gives (8). $\square$

4.2 Primitive-interval rigidity

Lemma 4.3 (A primitive radical element is a coordinate). *If $\theta \in R$ has zero constant term and nonzero linear part, an automorphism of R carries x_1 to θ . Consequently*

$$\text{ann}_R(\theta) = \theta R.$$

If η also has nonzero linear part and $\theta\eta = 0$, then $\eta = \theta u$ for some unit $u \in R$.

Proof. Every element of $\mathfrak{m} = (x_1, \dots, x_p)$ squares to zero: cross terms cancel in characteristic two and squares of nonconstant monomials vanish. Complete the linear part of θ to a basis of $\mathfrak{m}/\mathfrak{m}^2$, lift the other basis vectors, and send the generators to these p elements. Their squares vanish, so this defines an endomorphism of R . The induced map on the associated graded ring of the finite $\mathfrak{m}$ -adic filtration is an invertible linear change of generators. The endomorphism is therefore an automorphism.

Write $R = R' \oplus x_1 R'$ for $R' = \mathbb{F}_2[x_2, \dots, x_p]/(x_i^2)$. Multiplication by x_1 kills exactly $x_1 R'$, so $\text{ann}(x_1) = x_1 R$. Transport this identity by the automorphism. Thus $\theta\eta = 0$ gives $\eta = \theta u$. If u had zero constant term, the product would have zero linear part. Its constant term must therefore be one, so u is a unit. $\square$

Theorem 4.4 (Rigidity at multiplier one). *If $\kappa = 1$, then after scalar-unit cancellation C is isomorphic over R to one finite interval*

$$R \xrightarrow{\theta} R \xrightarrow{\theta} \dots \xrightarrow{\theta} R$$

on m , with θ of odd total-linear parity. A one-term interval is allowed and needs no coefficient. For every nonempty classical link completion of m ,

$$\dim H(\text{cl}(C)) = \dim \text{Kh}(L_m; \mathbb{F}_2). \quad (9)$$

In particular no knot-completion premise is needed here. Quantum homogeneity and genuine cube origin are not assumptions of this theorem.

Proof. After cancelling scalar units, the quiver of total-linear matrices has exactly one interval by Proposition 4.1. Hence its spaces are one-dimensional in consecutive degrees and zero elsewhere, and

its intervening binary arrows are one. The full differential has one coefficient θ_h between consecutive objects, and each θ_h has odd total-linear parity.

The chain condition gives $\theta_h \theta_{h+1} = 0$. Lemma 4.3 gives $\theta_{h+1} = \theta_h u_h$ with u_h a unit. Inductively each coefficient is a unit multiple of the first one. Successive rescalings of the rank-one free modules make every arrow equal to the same θ . This is a chain isomorphism over the full ring, not just its first jet.

Let D be the unreduced characteristic-two Khovanov complex of the classical link L_m , with commuting square-zero dot operators X_i . The vertical operator ν , which changes one circle label x to 1 and sums over circles, satisfies

$$\nu^2 = 0, \quad \nu d_D = d_D \nu, \quad \nu X_i + X_i \nu = 1.$$

These identities follow directly from multiplication and comultiplication in $\mathbb{F}_2[x]/(x^2)$ and hold for links as well as knots. Put $\Theta = \theta(X_1, \dots, X_p)$ and

$$U = \nu \Theta + \Theta \nu = \left(\sum_i \frac{\partial \theta}{\partial x_i} \right) (X_1, \dots, X_p).$$

The polynomial defining U has constant term one. Every zero-constant polynomial in commuting square-zero operators squares to zero, so $U^2 = 1$. The derivation $\partial = \sum_i \partial/\partial x_i$ satisfies $\partial^2 = 0$, giving $\nu U = U \nu$. All these operators commute with d_D . Thus $K = U \nu$ satisfies

$$K^2 = 0, \quad K \Theta + \Theta K = 1, \quad K d_D = d_D K.$$

For $E = \text{im } K = \ker K$, these relations give a direct sum of subcomplexes $D = E \oplus \Theta E \cong A_\epsilon \otimes E$, with Θ acting as multiplication by ϵ . The completed ℓ -term interval is the tensor product of E with the free dual-number interval J_ℓ . As a binary complex J_ℓ has dimension 2ℓ , differential rank $\ell - 1$, and homology dimension two. The field Künneth formula gives $\dim H(J_\ell \otimes E) = 2 \dim H(E) = \dim H(D)$. $\square$

The two ranks certify existence of the interval decomposition and rescaling; no such basis needs to be constructed by the decision query. The replacement asserts neither an isotopy of knots nor equality of open complexes or of original bigraded homology.

4.3 The boundary of the link extension

For $p = 2$, the two-term complex $R^2 \rightarrow R$ with row (x_1, x_2) has $t = 3$ and $\kappa = 2$. Completing to one circle identifies the variables and gives homology dimension four, as the knot formula predicts. For the two-component unlink the module is R itself. The image is (x_1, x_2) , which has dimension three. The total homology dimension is $12 - 2 \cdot 3 = 6$, whereas $\kappa \dim \text{Kh}(U_2) = 2 \cdot 4 = 8$. Thus the all-link equality cannot be extended to arbitrary κ . A three-term interval with coefficient $x_1 + x_2$ similarly has $\kappa = 3$ but completed rank four on the two-component unlink.

Raw survivor count alone is not a general algebraic decision rule: an interval with coefficient x_1 can have arbitrarily many objects and still have $\kappa = 1$. Such algebraic fixtures must not be presented as realized knot prefixes.

4.4 A flat-origin constraint on genuine minimal complexes

Proposition 4.5 (Vanishing of the minimal diagonal first jet). *Start from an ordinary undotted tangle cube over $\mathbb{F}_2$, apply delooping and scalar-unit cancellation, and retain any whole relative*

direct summand of the resulting minimal complex M . Every same-matching entry of its differential has total-linear dot parity zero. In particular a one-matching summand has $\kappa = t$.

Proof. In the dotted cobordism category, let ∂ remove one dot and sum over all choices. It is a derivation for composition and respects the relations in characteristic two. A one-dotted sphere differentiates to an undotted sphere, hence to zero. The derivative of the two-dot relation consists of two equal terms. The two dotted terms in neck-cutting also differentiate to two equal undotted surfaces. Their derivatives vanish. The undotted cube differential d therefore satisfies $\partial d = 0$.

For a chain isomorphism changing d to $D = FdF^{-1}$, differentiation of the inverse identity gives

$$\partial D = \Gamma D + D\Gamma, \quad \Gamma = (\partial F)F^{-1}.$$

Delooping is such an isomorphism. Gaussian cancellation is a chain isomorphism displaying a contractible disk as a direct summand; retain these disks formally until the end. The block of this equation on the remaining minimal summand still has the form $\partial D_M = \Gamma_M D_M + D_M \Gamma_M$, because the displayed differential has no off-block entries.

Recall the scalar-residue map ρ from the maintained radical analysis [20]. At a frontier of $2k$ endpoints, a dotted basis morphism $x_S : a \rightarrow b$ has nonnegative weight $k - c(a, b) + 2|S|$, where $c(a, b)$ counts the circles in the glued pair of matchings. Weight zero consists precisely of scalar identities. The Euler-characteristic composition rule makes weights additive in every nonzero valid composition. Projection to weight zero is therefore multiplicative; this is the map ρ . Minimality gives $\rho(D_M) = 0$, hence $\rho(\partial D_M) = 0$. On a same-matching entry this is exactly the sum of its singleton-dot coefficients. The residue on a different-matching entry is automatically zero. Restriction to a further whole summand preserves the same block argument. $\square$

This proposition needs genuine cube origin or a suitable connection identity. The generic query does not infer it from $Q^2 = 0$ or from a nonnegative survivor profile. It computes both ranks and remains valid on the broader algebraic class in Theorem 4.4. A future provenance-dependent optimization could omit the second rank; this release does not install that shortcut.

4.5 Decision control, costs, and the remaining structural question

Proposition 4.6 (Established rank-two unknot test over $\mathbb{F}_2$). *For a classical knot K ,*

$$\dim_{\mathbb{F}_2} \text{Kh}(K; \mathbb{F}_2) = 2 \iff K \text{ is the unknot.}$$

Every larger total unreduced rank is at least four.

Proof. Choose a basepoint and its dot map X on the ordinary complex D . The reduced complex can be taken as $\tilde{D} = \text{im } X$, up to its conventional quantum shift. The identities for ν above give $D = E \oplus XE$ for $E = \text{im } \nu$, and $X : E \rightarrow XE = \tilde{D}$ is an isomorphism with inverse ν . Both maps commute with the homological differential. Consequently

$$\dim_{\mathbb{F}_2} \text{Kh}(K; \mathbb{F}_2) = 2 \dim_{\mathbb{F}_2} \widetilde{\text{Kh}}(K; \mathbb{F}_2).$$

This is the characteristic-two reduced/unreduced splitting of Shumakovitch, Corollary 3.2.C [24].

The integral reduced cube is a bounded complex of finite free abelian groups. The universal coefficient theorem therefore gives

$$\dim_{\mathbb{Q}} \widetilde{\text{Kh}}(K; \mathbb{Q}) \leq \dim_{\mathbb{F}_2} \widetilde{\text{Kh}}(K; \mathbb{F}_2).$$

Kronheimer–Mrowka prove that the rational reduced rank is greater than one for every nontrivial knot: this is the rank consequence of their instanton spectral sequence and unknot-detection argument, specifically Corollary 1.3 and Proposition 1.4 with the accompanying discussion [10]. Hence a nontrivial knot has unreduced $\mathbb{F}_2$ rank at least four. Conversely the crossing-free unknot has complex $\mathbb{F}_2[x]/(x^2)$ with zero differential and rank two. All normalization shifts leave these total dimensions unchanged. $\square$

This detection theorem is an established topological input. Our contribution is the preservation of its total-rank observation under the new queries and replacements, not a new proof of Khovanov unknot detection.

For a classical link L with $c \geq 1$ components, $\dim_{\mathbb{F}_2} \text{Kh}(L; \mathbb{F}_2) \geq 2^c$. Indeed, write $J_L(q)$ for its graded Euler characteristic, normalized by $J_U(q) = q + q^{-1}$. The integral cube is free, so this Euler characteristic is unchanged by passage to $\mathbb{F}_2$. The skein relation $q^{-2}J_{L_+} - q^2J_{L_-} = (q^{-1} - q)J_{L_0}$ shows that crossing changes preserve $J_L(1)$. Changing crossings to obtain the c -component unlink gives $J_L(1) = 2^c$, and total homology dimension bounds the absolute Euler characteristic [7]. Homological or quantum normalization shifts do not alter this absolute bound. This argument is applied to the ordinary classical completion, whose total rank the replacement theorem preserves.

At each proper prefix the new driver fully allocates one crossing with cancellation disabled and finds whole differential components. A component with $\kappa = 0$ is discarded; one with $\kappa = 1$ can be replaced by one copy of its matching, with an arbitrary homological shift because only total rank is observed. Every nonempty classical completion of the latter contributes at least two to rank, and a completion with at least two link components contributes at least four. For $\kappa \geq 2$ the multiplier is used only after verifying that the actual completion is a classical knot. Contributions of independent components are added; a lower bound of four rejects the original knot.

Every replacement preserves total rank for every later classical completion. This statement is preserved by finite sequences of replacements; the theorem does not require quantum homogeneity of the intermediate rank-only model. The driver then uses ordinary scalar cancellation and continues its crossing order. At final closure it reports rank two as unknotted and larger rank as knotted, using Khovanov unknot detection.

The local optional allowance covers component traversal, polynomial extraction, binary work, completion geometry, and construction of replacement arrays. Arrays and reverse indices are prepared before a final check and one installation. Exhausting this allowance discards all uninstalled proposals from that stage, disables further observation, and resumes ordinary cancellation on the intact state. Global resource exhaustion propagates the ordinary scan limit and never becomes a knot verdict. The object ceiling includes temporary crossing allocation before observation.

For a queried block with $N = \sum_h n_h$ objects and total packed coefficient input length L , the query stores binary matrices using $O(N^2)$ bits up to index overhead and performs no cobordism composition. A conservative bit-complexity envelope is

$$\tilde{O}(L + N^3).$$

The tilde absorbs logarithmic factors and polynomial-size index arithmetic. Singleton monomials occupy packed bit positions $1, 2, 4, \dots$; reading only these avoids enumeration of dense irrelevant higher terms. The cost includes the given coefficient representation and does not assume it has polynomial size in p .

Let w be the maximum actual frontier size and g the largest number of crossings between successive one-object checkpoints of the executed scan, including its initial and terminal segments. The crossing ending a segment counts before observation or cancellation. A crossing creates at most

eight delooped children, so each segment has at most 8^g temporary objects. Coefficient spaces at width w have size $2^{O(w)}$. The explicit elimination, query, and geometry operations therefore give

$$T(n; g, w), S(n; g, w) \leq \text{poly}(n)2^{O(g+w)}$$

with the maintained polynomial ordering overhead. A one-object checkpoint can retain a large frontier, so w cannot be dropped as it can after a suitable diagram-splicing reset. Quasi-polynomial time follows if constructible orders guarantee $g + w = O(\log^2(n + 2))$. No such guarantee for arbitrary knots or unknots is established. Stages after observer exhaustion remain included in the actual recorded gap.

4.6 Finite evidence and interpretation

The discovery run uses seed 29002, accepting 180 knot braid diagrams from 591 attempts. Greedy and shuffled orders give 360 complete scans and 2898 stages. There are 117 eligible pre-Schur one-matching components, containing 633 objects and 237 scalar pairs. All are nonsingleton and contain scalar units. There are 96 cases with $t = \kappa = 1$ and 21 with $t = \kappa \geq 2$. All 117 pre/post-cancellation comparisons agree. None of 8183 inspected minimal same-matching entries has odd total-linear parity, consistently with the flat-origin proposition.

These counts establish real eligibility before Schur updates, rather than only synthetic interval fixtures. They do not establish a timing gain; queries and traversal incur overhead. Paired timing data must be read separately. In particular no $t > \kappa$ primitive-interval compression occurs in this actual-diagram corpus.

The tests compare decisions with an independent complete crossing cube, test continued scans after replacement and exhausted local budgets, and evaluate polynomial complexes with scalar units and nonlinear terms under random invertible changes of basis. They retain explicit link-completion counterexamples and an invalid polynomial complex whose first-jet identities still hold, documenting the precise audit scope.

Further research questions. Can a cheaply maintained cube-origin connection certificate eliminate the second rank throughout the decision scanner? Can small scalar components be queried without traversing the whole differential graph while still certifying absence of external attachments? Which geometric classes admit polylogarithmic one-object checkpoint gaps and frontier widths? Can the flat-origin constraint be lifted beyond scalar residue to restrict higher jets of mixed-matching minimal complexes? Can hard reduced knot families be found where avoided cobordism work outweighs observer overhead?

5 Reverse-suffix response compilation with singular interiors

The preceding fixed-suffix response construction leaves an avoidable source of repeated work: after the scan advances, the next suffix is usually rebuilt and its entire interior is eliminated again. We give a different construction which prepares *all* suffix responses in reverse order. Its quantitative gain is a theorem about response compilation on diagrams of bounded frontier. It supplies neither a universal frontier bound nor a bound on the number of Khovanov components. The new implementation is an experimental module; no recognizer driver dispatches to it automatically.

This construction belongs to the established study of elimination on graphs of small width. In particular, Fürer, Hoppen and Trevisan give an $O(k^2(m + n))$ rank and determinant algorithm over

arbitrary fields when a suitable width- k decomposition is supplied [5]. We do not claim a new general small-width Gaussian-elimination bound. The contribution here is the all-suffix determinant response interface, its exact compatibility with cut-face identifications, and a singular-safe implementation whose updates can be reused across actual stages of the unknot scanner.

5.1 The admissible graph filtration

Fix a checkerboard coloring of the source diagram once for the entire filtration. In this section, α pairs the two darts of each diagram edge, and ρ rotates the four darts of each crossing counterclockwise. For a suffix, its common Tait vertices are black cut-face fragments, and its edges are the signed crossing edges. Build suffixes in reverse: start with the last crossing, then insert the preceding crossing, and continue. A crossing initially contributes four new face fragments, of which exactly two are black. Whenever a new edge joins an already present dart, the face permutation adds the two connections

$$d \longleftrightarrow \rho(\alpha(d)), \quad \alpha(d) \longleftrightarrow \rho(d).$$

These connections merge fragments. They never split a fragment.

A closed fragment is already a complete cycle of the partial face permutation. All of its incoming and outgoing permutation arrows are present; no subsequent insertion can touch it. Consequently every later identification occurs among open fragments or fragments just introduced. Every later signed edge has endpoints among those same fragments. This is the essential admissibility condition for retaining an eliminated interior. It would be false for arbitrary changes of a diagram or for forward deletion of an already eliminated crossing.

Keep one distinguished black fragment as a grounding anchor, even if it closes. When it merges with another open fragment, follow the merged fragment. This leaves at least one retained coordinate at every stage. If the suffix has w boundary darts, at most w black fragments are open, so the number b of retained black coordinates obeys

$$1 \leq b \leq w + 1.$$

The extra coordinate is unnecessary when the anchor is already open. The anchor represents an actual graph vertex; it is not an isolated vertex added to the graph.

5.2 A response preserved under arbitrary later attachments

Work over a field $\mathbb{F}$. Starting from a symmetric signed Laplacian, eliminate only interior coordinates by nonsingular symmetric principal pivots. A nonzero diagonal gives a one-coordinate pivot. If all remaining interior diagonals are zero but an off-diagonal entry c is nonzero, the principal block

$$\begin{pmatrix} 0 & c \\ c & 0 \end{pmatrix}$$

is invertible and gives a two-coordinate pivot. If neither kind exists, the residual interior block is identically zero. These alternatives hold in characteristic two as well as in odd characteristic.

The resulting response consists of a nonzero scalar δ and

$$M = \begin{pmatrix} 0_r & R \\ R^\top & S \end{pmatrix}, \quad R \in \mathbb{F}^{r \times b}, \quad S = S^\top \in \mathbb{F}^{b \times b}.$$

Here δ is the product of all eliminated principal determinants. For any additional graph attached only at the retained coordinates, and for any identifications among those coordinates, a grounded cofactor of the augmented original graph equals δ times the corresponding grounded determinant obtained from M . To see this, add the exterior matrix in its own coordinates before eliminating. Every eliminated principal block is unchanged, its coupling to new exterior coordinates is zero, and block Gaussian elimination therefore gives the same Schur complement and determinant factor. Terminal identification is the congruence $M \mapsto T^T M T$ with T equal to the identity on interior coordinates. This congruence commutes with the same interior elimination. Grounding a retained terminal also commutes with it.

No invertibility assumption is imposed on the full interior block. Discarding 0_r and R would be unsound. For example, over $\mathbb{F}_2$ the path 0–1–2, with terminal endpoints, has the single interior diagonal $2 = 0$, but its tree cofactor is one. Retaining its one-dimensional radical permits a later two-coordinate pivot when the path is extended; treating the zero interior pivot as a zero answer would lose a valid response.

5.3 The absorbing-zero lemma

Lemma 5.1 (Absorbing zero). *Suppose a residual response has $r \geq b$. Every grounded cofactor obtained by any future admissible graph attachments and terminal identifications is zero over $\mathbb{F}$.*

Proof. Schur complementation of a zero-row-sum matrix still has zero row sums. Indeed, if $L = \begin{pmatrix} P & B \\ B^T & C \end{pmatrix}$ annihilates the all-ones vector, then $P\mathbf{1} + B\mathbf{1} = 0$ and $B^T\mathbf{1} + C\mathbf{1} = 0$, whence $(C - B^T P^{-1} B)\mathbf{1} = 0$. The same is true after terminal congruences and addition of signed Laplacian edges. Since the first r rows of M have zero interior block, $R\mathbf{1}_b = 0$. Thus $\text{rank } R \leq b - 1 < r$, and there is a nonzero $u \in \ker R^T$. The vector $(u, 0_b)$ is in the kernel of M . Its support lies wholly in permanent interior coordinates, which future attachments and identifications never touch. Extending this vector by zeros gives a nonzero nullvector of every future grounded matrix, since the grounded coordinate is retained rather than interior. The determinant is therefore zero. $\square$

The lemma permits an *absorbing* zero response: all numerical matrix entries may be discarded, although boundary geometry must still be maintained. The module records this zero separately from a missing or interrupted computation. This is a field-specific zero. A zero modulo one prime never proves that an integer determinant is zero.

The sufficient test $r \geq b$ is intentionally cheap. A rank-deficient R can yield a permanent nullvector even when $r < b$; recognizing all such cases would require another rank calculation. The additional test is unnecessary for the claimed size bound. Every carried nonzero response has

$$r \leq b - 1, \quad \dim M \leq 2b - 1.$$

5.4 Why every crossing needs only constantly many pivots

Consider one admissible update. Aggregate the old terminal rows and columns under all identifications, insert the new crossing edge, and move the newly closed black fragments into the interior. Let their number be u . The old radical block remains zero, so the interior matrix now has the form

$$A = \begin{pmatrix} 0_r & C \\ C^T & D \end{pmatrix}, \quad D \in \mathbb{F}^{u \times u}.$$

The first r rows are supported on u columns; the final u rows contribute rank at most u . Hence

$$\text{rank } A \leq 2u.$$

Symmetric nonsingular principal elimination removes exactly its pivot rank and leaves an identically zero interior block. It therefore eliminates at most $2u$ coordinates, using at most $2u$ pivots.

Inserting one crossing creates at most four physical internal edges, each supplying one black face connection. Each newly closed black fragment must contain a newly supplied connection, so $u \leq 4$. Retaining the anchor can only reduce u . Thus there are at most eight one- or two-coordinate pivots per crossing, including exceptional characteristics in which pivot profiles change. No probabilistic genericity or numerical stability assumption is used.

5.5 All-stage complexity theorem

Theorem 5.2 (All-stage response compilation). *Let a spherical n -crossing source diagram have a prescribed crossing order whose largest dart frontier is W . Over any fixed field, all nonempty suffix determinant responses can be compiled with $O(n(W+1)^2)$ field operations. Keeping all snapshots stores $O(n(W+1)^2)$ field elements; streaming the snapshots uses $O((W+1)^2)$ field elements in addition to $O(n)$ geometric storage. An arbitrary checked boundary-matching query costs $O((W+1)^3)$ field operations and $O(W\alpha(W) + W)$ combinatorial operations. These bounds include singular interiors.*

Proof. For a nonzero response, $b \leq W+1$ and $r \leq b-1$. At most two black vertices are introduced in the next update. Identifications do not increase matrix dimension. Matrix construction, terminal aggregation, and each symmetric pivot therefore cost $O((W+1)^2)$ field operations. The preceding rank argument bounds the number of pivots per crossing by an absolute constant. A scan for an available diagonal or off-diagonal pivot also examines at most $O((W+1)^2)$ entries and is repeated only constantly often. If the absorbing-zero condition occurs, subsequent arithmetic is cheaper. Summing over n crossings proves the arithmetic claim. Copying a finished immutable response has the same quadratic bound.

Original edge pairing, source faces, checkerboard coloring, and the insertion sequence are prepared in $O(n)$ graph work apart from disjoint-set costs. Union by size with path compression handles $O(n)$ unions and $O(n + \sum_s w_s)$ finds. Snapshot construction visits only the current boundary, sorting its labels and roots once. Its total combinatorial work is

$$O\left((n + \sum_s w_s)\alpha(n) + \sum_s w_s \log(w_s + 2)\right).$$

It never retraverses the whole suffix. This separate statement avoids silently treating union-find costs or index bit lengths as field operations. The geometric storage consists of the source darts, two union-find structures, the active-crossing flags and the boundary dictionary, all of size $O(n)$; a snapshot retains $O(W)$ labels.

For a query, boundary unions reconstruct exactly the completed open faces and projection components. The inherited coloring is checked, and the sphere equation $F = n_s + 2k$ is verified. Closed fragments and closed projection components contribute their stored counts. There are at most W open black fragments. Their quotient partition, together with the fixed anchor when closed, grounds a matrix of dimension at most $2b - 2 = O(W+1)$. Dense field elimination costs the stated cubic amount. $\square$

Since $W \leq 4n$, the construction also gives an unconditional $O(n^3)$ field-operation bound for *all-stage response compilation*, where independent dense cubic setup at every stage gives the coarser $O(n^4)$ bound. For $W = O(1)$ the new arithmetic bound is linear; for $W = O(\log^c n)$ it is $O(n \log^{2c} n)$. None of these statements bounds the number of observations or the work of the relative-complex scanner that requests them.

For a variable prime p , multiplication costs and inversions must be restored to the field-operation bound. With $\ell = \lceil \log_2 p \rceil$, the construction uses $O(n(W+1)^2)$ operations on $O(\ell)$ -bit field elements and only $O(n)$ inversions. The implementation validates primes below 2^{31} , including two, so its individual field elements have a fixed machine-word bound. More primes multiply the compilation cost; the theorem does not claim constant-cost reconstruction of an arbitrarily large integer cofactor.

5.6 Implementation and integration boundary

The new file `fastunknot/streaming_response.py` supplies immutable `StreamingResponse` and `SuffixResponse` objects, an iterator over suffixes in reverse stage order, and a convenience function returning every snapshot indexed by forward prefix stage. The generic arithmetic API rejects new edges whose endpoints are no longer retained terminals. The geometric compiler additionally checks every black-fragment merge against its live terminal set, so an accidental identification of an eliminated face is an explicit error.

Snapshot completion queries check exact boundary coverage, inherited color compatibility, and spherical genus. A disconnected projection has raw reduced Jones value zero at i . A connected query uses the phase $(-i)^{B+v-1}$, where $B = \sum_e b_e$: the crossing bit is $b_e = 1$ for black sectors $[0, 2]$ and $b_e = 0$ for black sectors $[1, 3]$, and the edge weight is $(-1)^{b_e}$. This sum retains contributions of Laplacian loops. The number v is the actual completed black-face count. The anchor is included in this count exactly once. Modulo a prime the result is stored as a formal Gaussian pair; it does not require choosing a square root of -1 in the field.

All response updates and queries poll an optional cancellation/work callback. An exception propagates before a partially completed snapshot is published. Existing snapshots cannot be modified by an update. The constructor's source validation is also interruptible. There is no new default setting, fallback-policy change or positive unknot claim.

The iterator accepts an optional frontier allowance and stops before a response at an over-wide stage is built. The all-snapshot convenience function also accepts a retained-field-element allowance, charging the sum of squared stored response dimensions before retaining each completed snapshot. Exhaustion raises a distinct `ResponseBudget` exception; the convenience function returns no partial compiled tuple. The storage allowance covers retained field entries, not source metadata or a current update's temporary matrix. It is not a hard process-memory limit.

The primary integration opportunity is an observer that expects to query many stages of a fixed order. It can prepare responses once and query the corresponding stage without repeating interior elimination. Conversely, a scan that is likely to terminate after one inexpensive early observation should not automatically prepare every suffix. An adaptive dispatcher must account for the all-stage setup cost. No such dispatcher or end-to-end speedup is claimed here.

5.7 Independent checks and measurements

The reproducible integer-cofactor audit accepts 53 knot braid diagrams from 240 seeded attempts. It considers every one of their 610 suffixes under a recorded shuffled order, over the primes 2, 3, 5, 101. There are 2440 field responses and 19520 independent quotient-cofactor comparisons, all agreeing.

The sample includes 493 singular responses, 19 absorbing-zero responses, and 72 streams that perform a two-coordinate pivot. The largest retained radical has dimension four. The reference reconstructs complete cut-face graphs by graph components and computes integer Bareiss cofactors; it does not reuse the incremental fragment union-find or the field-response matrices.

Eight focused tests additionally exercise actual scanner matchings and their raw Gaussian phases, mirrors, randomly generated graph insertion streams with identifications and forgetting, the characteristic-two singular path example, protected zero responses, immutable updates, source and mid-update cancellation, and the explicit frontier/storage allowances. No first-jet or recognition driver is imported by the new response compiler.

The timing family is the natural crossing order of the positive three-strand braid $(\sigma_1\sigma_2)^{n/2}$ for the listed n . The closures are knots since $n/2$ is coprime to three. Both arms prepare every suffix of the same diagram. The dense control independently rebuilds each cut-face graph and uses the same symmetric pivot arithmetic; the streaming arm retains elimination between suffixes. Original coloring is computed once per arm. Source preparation, geometry and all response setups are included. One excluded warmup checks five arbitrary quotient queries per suffix. Shuffled A/A/B rounds include two identical dense controls. The first four rows use seven rounds; the last uses three.

n	Dense setup (ms)	Streaming (ms)	Ratio	Dense A/A
16	1.308	0.522	2.51	1.076
32	5.584	1.123	4.97	1.155
64	27.180	2.193	12.39	0.993
128	267.265	5.241	50.99	0.992
256	3615.101	14.061	257.10	0.906

The large ratios reflect substantial interior reuse. At $n = 256$, 253 of 256 stages have a nontrivial interior and 84 have a nonzero retained radical. The dense common graph reaches 129 black vertices; the largest matrix constructed by streaming has dimension five. Each dense arm performs 23789518 symmetric elimination updates across all suffixes, compared with 928 for streaming. At all natural-order sizes the largest dart frontier is six. Compiling all 8192 suffixes of the corresponding larger braid takes 0.4155 seconds in one additional streaming-only run, performs 30024 elimination updates, and stores 92825 field elements. That run has no dense timed comparator.

Recorded shuffled orders at 16, 32 and 64 crossings give smaller ratios of 1.69, 2.50 and 3.73. This sensitivity is expected: a poor order increases the actual retained boundary. Small-instance timings and the 256-crossing dense controls exhibit visible noise; the table reports their controls rather than treating milliseconds as exact. The operation-count theorem is independent of these wall-clock ratios.

These knots are readily decided by existing recognizer filters. The measurements establish a response-compilation improvement, not a whole-recognizer speedup or a difficult family for unknot recognition. Their significance relative to earlier boundary benchmarks is that substantial, sometimes singular interiors are eliminated once across genuinely changing suffixes. All raw input words, orders, timings, counts and source hashes are retained in `fast/results/streaming_response*.json`; the source hashes in all three final JSON files match the delivered response module.

5.8 Further questions specific to this construction

1. Can a cost-based observer choose a prefix of the reverse compilation or a collection of checkpoints that gives a competitive bound relative to the better of independent setup and full precompilation?

2. Can the same face filtration compile the two Euler boundary path summaries and determinant response in one pass, so stage geometry and cancellation checks are shared?
3. Can a separator tree replace a linear crossing order while preserving singular residual information, giving local updates after a small diagram change? The present no-future-interior-identification contract must be replaced by a proved tree-boundary contract.
4. Which real observer traces contain enough nontrivial interior work at enough stages for this compilation theorem to improve complete recognition? Easy torus-braid families isolate the arithmetic gain but cannot answer this question.
5. Can boundary response snapshots be shared across several primes or several crossing orders without assuming a common pivot profile? Exceptional primes require exact handling of changing radicals.

6 How the results could form a quasi-polynomial portfolio

6.1 Three sufficient regimes

The previous sections give different, explicitly checkable sufficient conditions. They should be combined by a logical decision policy and a complete cost account, rather than by adding their most favorable exponents informally.

Corollary 6.1 (Explicit sufficient regimes). *For diagrams with n crossings, any one of the following conditions gives a deterministic $n^{O(\log n)}$ recognition method, including its stated construction cost:*

- (i) *A verified presentation can be constructed in polynomial time with $r = O(\log n)$ and explicit relator length $N = n^{O(1)}$.*
- (ii) *A verified word circuit of polynomial description length and a cut can be constructed in polynomial time with $(r + t) \log(d + 2) = O((\log n)^2)$.*
- (iii) *An order for the first-jet decision scan can be constructed in polynomial time whose actual one-object gap and frontier obey $g + w = O((\log n)^2)$, with all temporary allocation and the terminal segment included.*

No condition is asserted to hold on arbitrary diagrams. In (i) and (ii), the decision method is the theoretical exact real procedure, not an asymptotic guarantee for the supplied practical solver.

Proof. In (i), the logarithm of the nonpolynomial factor is $O(r \log(2 + N/r)) = O((\log n)^2)$. Condition (ii) is precisely the exponent of Theorem 3.15. Condition (iii) follows from the scan bound in Section 4. In each case the additional polynomial factors are absorbed. $\square$

The supplied-diagram condition $b(D) = O(\log n)$ is a concrete instance of (i) whose construction is already implemented. Fully retaining a word circuit with $r + s = O(\log^2 n)$ is a concrete instance of (ii). Conditions (i) and (iii) describe different geometric and algebraic phenomena: neither a small bridge invariant nor a small matrix response may be substituted for the measured parameters in their proofs.

6.2 Certified reductions and fair execution

Proposition 6.2 (A portfolio sufficient condition). *Suppose a polynomial-time verified reduction converts an input diagram into polynomially many knot instances of total polynomial size, with the original knot trivial if and only if every resulting knot is trivial. For each instance suppose at least one member of a fixed finite collection of sound algorithms terminates within $n^{O(\log n)}$ steps, with uniform constants. Then a fair execution of that collection decides the original input in $n^{O(\log n)}$ time. Every proposed certificate must be checked within the corresponding cost, and the instance-size bound refers to the actual encoded inputs.*

Proof. Simulate the fixed number of algorithms on each instance in round-robin instruction blocks. Stop that instance when one supplies a checked verdict. If an algorithm takes C steps, the simulation gives it those steps with only a fixed-factor scheduling overhead in a standard multi-workspace machine model; a conventional bit-machine simulation adds at most polynomial or logarithmic overhead, harmless here. Sum over polynomially many instances. Soundness and the verified reduction justify the conjunction of their unknot verdicts. A single checked knotted instance permits earlier rejection. $\square$

This elementary proposition is not a new structural theorem. It locates the missing proof: every residual input must be covered by at least one bounded regime. Finite budgets, empirical dispatch thresholds, and a collection of examples do not establish that covering statement. A bounded practical solver that can return unknown is a sound portfolio member, but does not by itself supply the terminating algorithm required by the proposition.

For a scan making q boundary observations over a primes, the new response contribution has the schematic arithmetic cost

$$O(an(W + 1)^2 + aq(W + 1)^3).$$

Source geometry and index arithmetic have the separately stated costs in Section 5. If a or q is large, that must be paid. The number of relative objects and cobordism compositions remains in the rest of the scan's cost. This is why the response theorem is an ingredient within a recognition algorithm, not a complete recognizer to place by itself in the portfolio minimum.

6.3 Correct integration contracts

The delivered additions expose narrow interfaces with explicit verdict semantics. A generic group presentation produces a representation query; only a formula derived from a validated knot diagram is promoted to a knot verdict. Defining-generator elimination preserves the surviving positive meridians, which permits the shared scalar coordinate and the meridional inequality. Arbitrary future Whitehead transformations must use the general commutator test unless conjugacy provenance is retained.

The first-jet observer accepts whole components of the actual differential graph. It does not treat a selected submatrix as a direct summand. Local exhaustion discards uninstalled proposals, disables observation, and resumes ordinary elimination; global exhaustion propagates. Replaced complexes support total-rank decisions only. A homology-reporting API must continue to compute the original graded complex.

The response module returns a field-specific cofactor response. Its absorbing zero is proved for every admissible continuation, but only in that field. A zero residue is not an integer zero certificate, and compatibility with an unknot determinant is not a positive recognition certificate. Completion geometry, signs, loop phases, and the anchor remain part of the exact query.

New interface	Definite output means	Inconclusive or excluded behavior
<code>su2.solve_formula</code>	Exact nonabelian representation exists, or exact infeasibility for its input formula	Solver unknown, timeout, missing solver, or compilation allowance.
<code>su2.su2_decide</code>	Knot verdict derived from validated diagram provenance	Practical solver trust is explicit; generic external presentations are not accepted as diagram provenance.
<code>first_jet_profile</code>	Two exact ranks of a whole one-matching component	No inference from first-jet identities alone that an arbitrary polynomial matrix squares to zero.
<code>first_jet_khovanov_decide</code>	Total-rank decision preserved through its replacements	Original bigraded homology is outside the API contract.
<code>compile_suffix_responses</code>	Every prepared suffix has an immutable exact response modulo the supplied prime	No automatic driver dispatch or general recognition-time claim.

The default backend selection is unchanged. The new modules are optional research interfaces because the measurements do not yet justify a uniform production dispatch policy. One unrelated existing subprocess polling defect is corrected by integrating a previously developed fix; its provenance and regression test are documented in [Appendix A](#).

7 Implementation evidence and controlled comparisons

7.1 Verification scope

The integrated implementation passes 728 unit tests, with three skips for unavailable optional dependencies, in the recorded Python 3.12.14 environment. The run includes all 31 newly added tests: ten for exact quaternion and presentation compilation, thirteen for first jets and continued rank-decision scans, and eight for streaming responses. After the full run, an additional general-presentation API boundary was tightened so callers cannot request the meridional shortcut without validated diagram construction; the ten $SU(2)$ tests were rerun and passed, including a same-trace cyclic counterexample. The final archive retains the logs and source manifest rather than relying on a test count in a historical README.

The test oracles are intentionally different computations. Streaming responses are compared with integer Bareiss cofactors from independently reconstructed complete graphs. First-jet decisions are compared with an independent full crossing cube, as well as the maintained exact scanner. Representation queries are compared with a separate reduced Khovanov cube and finite-group homomorphism counts. These checks test implementation behavior on finite inputs; the theorems establish the universal statements.

Recorded check	Coverage	Result and qualification
Integrated test suite	728 tests, including 31 new tests	Passed; three optional-dependency skips.
First-jet discovery	360 scans, 2898 stages; 117 eligible components	All pre/post-elimination profiles agree; all observed $t = \kappa$.
First-jet independent cube	96 comparisons on 32 sampled knots	No decision disagreement; continued replacements and budget fallback included.
Streaming integer oracle	53 diagrams, 610 suffixes, 19520 comparisons	No mismatch; singular and characteristic-two cases included.
SU(2) short-budget audit	96 queries on 48 sampled entries, 38 distinct PD arrays	90 complete and correct, six inconclusive.
Finite-group presentation audit	36 sampled entries, 26 distinct PD arrays	Equal S_3 homomorphism counts for all three presentations.

7.2 First-jet observer: eligibility and timing are different results

The discovery experiment in Section 4 establishes that the pre-elimination test encounters eligible components in actual diagrams. Its 117 components contain 633 objects and 237 scalar pairs. All contain scalar units, so they lie outside the old radical-only observer’s precondition at that moment. This does not by itself imply a speedup: traversing the whole graph and computing ranks also costs work.

The paired benchmark therefore compares six raw-scanner arms on twelve actual diagrams: the ordinary scanner, an identical control, the full first-jet scanner, an observe-only variant, a variant without completion geometry, and a zero-observation-budget fallback. Every arm validates the same supplied PD and uses the same crossing order, no shape cache, a three-second deadline and a 50,000-object ceiling. Timings include validation, scanner construction, observation, and elimination. They exclude imports, input construction, order selection, and one warmup. Front-end recognizer filters are bypassed. Seven shuffled rounds are recorded; small cases batch repeated calls as specified by the script. All arms finish in every recorded case.

Input	n	Standard ms	Jet ms	Paired ratio	A/A	Repl.
Trefoil	3	0.336	0.364	0.922	0.954	0
Conway	11	3.362	4.605	0.795	1.009	0
Kinoshita–Terasaka	11	4.625	6.410	0.761	1.018	0
Hard unknot fixture	8	0.780	0.840	0.893	1.065	0
Scrambled grid unknot	6	0.363	0.682	0.525	1.033	3
Five-strand stress braid	36	301.156	483.852	0.663	0.964	0
Conway connected sum	22	52.323	15.007	3.367	1.009	0
Coxeter unknot	11	0.334	1.077	0.310	0.989	10
Coxeter unknot	35	1.147	5.864	0.192	0.895	34
Trefoil with 8 cancelling pairs	19	1.795	2.110	0.848	1.015	0
Trefoil with 24 cancelling pairs	51	4.569	5.651	0.750	1.008	0
$T(3, 5)$	10	2.440	2.702	0.895	0.988	0

Table 2: All twelve first-jet benchmark cases. Times are medians; “paired ratio” is the median of the seven within-round standard/jet ratios and therefore need not equal the ratio of the two displayed medians. A/A is the corresponding standard/control statistic. Repl. counts installed multiplier-one replacements. Ratios below one mean the first-jet arm is slower.

The clear gain is the connected sum of two Conway diagrams. At crossing 11 of 22 it has a 975-object whole one-matching block with scalar rank 471 and $t = \kappa = 33$. The actual matching completion is a knot, so the rank lower bound is 66 and the scan rejects immediately. Counted cobordism compositions fall from 3324 to 39. No replacement occurs. The observe-only arm has paired ratio 3.473, confirming that the gain is early rejection. Disabling completion geometry removes this certificate and has ratio 0.569.

The three cases that actually install replacements are slower, with ratios 0.525, 0.310, and 0.192. Their ordinary scans already perform zero counted cobordism compositions, so there is little expensive work for the first-jet query to avoid. The remaining full-jet ratios range from about 0.663 to 0.922. These unfavorable results are consequential: the implementation is useful as an exact experimental interface, but the data do not justify a default global replacement policy. The Conway-sum case is also decomposable and does not establish a whole-pipeline improvement over the repository’s existing structural reductions.

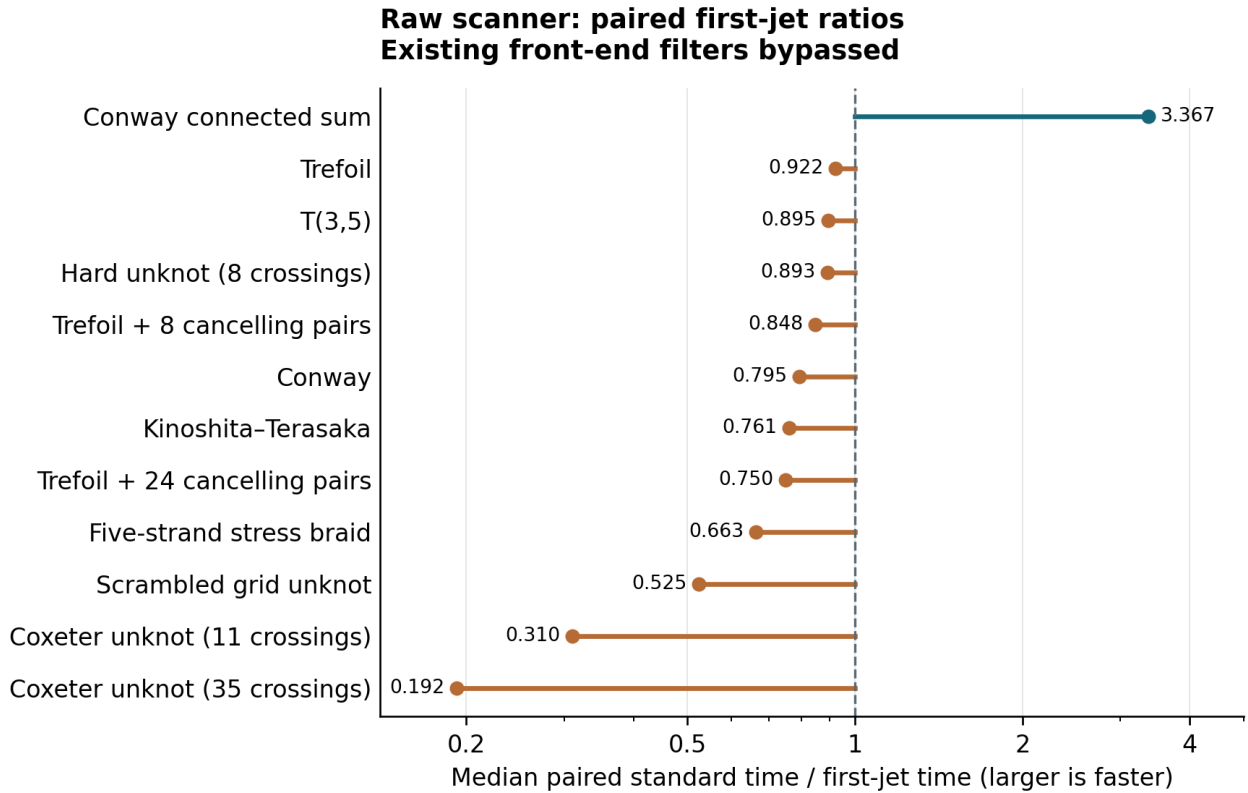


Figure 1: All first-jet paired ratios from Table 2. The dashed line is equal runtime. The early Conway-sum obstruction is the only improvement in this raw-scanner corpus; replacement cases regress.

7.3 SU(2) compilation and decision experiments

The recorded experiments test the exact formula construction and illustrate the practical degree-variable tradeoff with the exact NLSAT method[4, 6]. They do not measure the running time of the theoretical real-decision algorithm. The complete records are `results/su2_audit.json` and `results/su2_benchmark.json`; the corresponding scripts are `su2_research/audit_su2.py` and

su2_research/benchmark_su2.py.

Finite correctness audit. With seed 20261008, the audit sampled 48 accepted one-component braid closures on two to four strands, using words of one to seven letters. Every third accepted diagram was mirrored. An independently implemented reduced $\mathbb{F}_2$ Khovanov cube classified 39 samples as unknots and nine as knotted. Bridge compilation completed 42 of 48 queries: 36 unknot decisions and six knottedness decisions. Its six timeouts comprised three samples of each oracle class. Defining-generator Tietze elimination followed by compilation completed all 48 queries. Thus 90 of 96 queries were definitive, with no disagreement among those 90. The audit identifies its exact backend as Z3 5.1.0.

These are 48 sampled entries, containing 38 distinct stored PD arrays, and do not constitute 48 distinct knot types or an independent test-set estimate of general success probability. The saved PD arrays specify the actual inputs: the stored source braid words precede the optional mirroring step. The seed and script reproduce that step. For the 36 sampled entries with at most five crossings, comprising 26 distinct PD arrays, exhaustive counts of homomorphisms to S_3 agreed for the Wirtinger, bridge, and simplified presentations. This checks concrete relation-order and substitution behavior; agreement for one finite target group does not establish presentation equivalence in general. The Tietze and bridge proofs supply that justification.

The same audit checked the prime-pretzel family at $m = 3, 5, 7, 11, 21, 51$, hence at 9, 15, 21, 33, 63, 153 crossings. The computed Fox 3-coloring dimensions were respectively 3, 5, 7, 11, 21, 51, as predicted by Proposition 3.19. These finite checks support the construction and indexing; the proposition’s proof supplies the statement for all odd m .

Presentation choice and peak degree. The benchmark makes three repetitions per presentation on seven fixed examples, with a requested solver allowance of 250 milliseconds. Table 3 reports representative cases. Here k is the actual number of real variables after the safe gauge and meridional trace sharing, and d_{eq} is the maximum degree of an individual compiled equality before squaring. The implementation sends the separate equations to NLSAT in these measurements. Its sum-of-squares aggregate is available separately for the constant-atom theoretical bound.

Input	Presentation	k	d_{eq}	Compile ms	Query ms	Result
Figure-eight	Wirtinger	10	4	0.702	273.510	I
	Bridge	10	2	0.416	270.820	I
	Tietze	4	10	1.716	26.103	K
$T(2, 9)$	Wirtinger	25	4	1.866	283.346	I
	Bridge	25	2	0.972	285.034	I
	Tietze	4	18	11.411	72.254	K
Coxeter unknot, $n = 9$	Wirtinger	25	4	1.519	280.035	I
	Bridge	2	2	0.323	0.983	U
	Tietze	2	2	0.319	1.232	U

Table 3: Recorded medians of three repetitions, in milliseconds. K: nonabelian representation; U: no nonabelian representation; I: inconclusive timeout. All three repetitions in each row had the displayed status. The Coxeter example is the closure of $\sigma_1 \cdots \sigma_9 \in B_{10}$. Timeout durations are reported as censored runs and are not completion times.

On $T(2, 9)$, Tietze elimination reduced k from 25 to four while raising d_{eq} to 18, with aggregate degree bound 36. Its compilation took longer, but its query completed within the requested allowance while the other two formulations did not. This demonstrates why generator elimination must be

assessed together with degree and support growth. It gives no finite speedup ratio against a censored run. Across all 63 benchmark queries, 39 were definitive and 24 inconclusive.

Compressed powers. A separate construction-only experiment uses the general presentation $\langle x_1, x_2 \mid x_1^{2^b} \rangle$, represented by b squaring gates. It has no knot-group provenance and was not assigned a knot verdict. Table 4 shows that the compiler retained the circuit, producing constant-degree equations without expanding 2^b letters. The number of stored terms sums the supports of the individual equations and the nonabelian polynomial; it is not the support of a fully collected sum-of-squares aggregate.

b	Real variables	Stored terms	d_{eq}	Compile ms
8	37	97	2	0.260
32	133	361	2	0.815
128	517	1417	2	3.160
512	2053	5641	2	13.622
2048	8197	22537	2	54.236

Table 4: Three-run compiler medians for an already constructed word circuit. Expanded word length is 2^b ; aggregate degree bound is four throughout. These measurements exclude WordArena construction and make no claim about solver performance on the resulting formulas.

Interpretation and trust boundary. Both scripts enable a trace-zero witness probe before the unrestricted query. A satisfying assignment to this restricted query is valid for the original formula; an unsatisfiable or unknown probe is discarded by this implementation. The known completeness theorem for verified knot meridians, Theorem 3.6, would permit a stronger knot-specific policy, but these recorded runs do not implement it. All 15 successful knottedness queries in the audit used this probe. Their exact algebraic model expressions were retained and checked against the assertions by Z3 itself. This remains a trusted-solver computation, not an independently checked proof certificate. Neither a trace-zero failure nor any timeout is accepted as an unknot decision.

The timing allowances are soft: the largest recorded audit call took 375.926 milliseconds with a requested 300-millisecond allowance, and the largest benchmark query took 288.326 milliseconds with a requested 250-millisecond allowance. The benchmark’s reported query time includes formula parsing and solver checks, but excludes the Z3 import and subsequent model extraction and validation. Its compiler times exclude construction of the input diagram. The audit’s per-call times cover compilation and the solver wrapper.

The identical repeated compiler controls also show appreciable timing variation: across 63 pairs, the median first-run/control ratio was 1.234, with range 0.609–5.565. Since the control always runs second, this includes order and warm-up effects. With three repetitions, fixed case order, and a small convenience sample, the data justify neither a statistical speedup claim nor an extrapolated asymptotic exponent. They document correct completed queries and concrete representation tradeoffs. Renegar’s theoretical complexity and this optional NLSAT implementation remain separate statements.

7.4 Interpreting the response measurements

Section 5 reports the complete all-suffix setup experiment because its arithmetic and geometry scopes belong together. At 256 crossings, dense rebuilding takes 3615.101 ms and streaming takes

14.061 ms, a recorded ratio of 257.10; the identical dense control ratio is 0.906. The new arithmetic theorem, the reduction from 23,789,518 to 928 elimination updates, and the small carried matrices explain the direction of the gain without depending on an exact wall-clock ratio.

The input family is easy for the recognition front end. The streaming-only 8192-crossing run demonstrates capacity for many evolving suffixes, not difficulty of those knot types. A future integration experiment must retain inputs on which the recognizer actually demands many such responses, and include the setup cost even if an early certificate ends the scan. The present release therefore makes no default-recognizer speedup claim from this experiment.

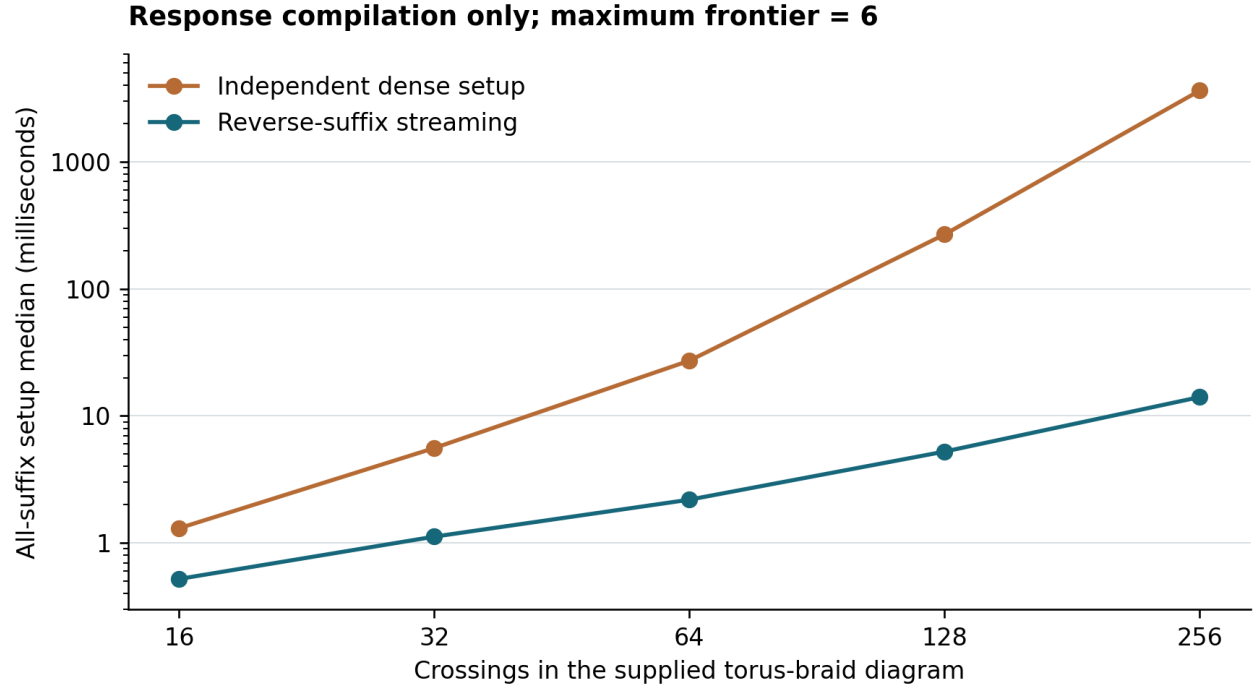


Figure 2: Recorded all-suffix setup medians at frontier six, with logarithmic axes. These are easy torus-braid inputs and subroutine timings. The plot shows the recorded points only; it does not fit an asymptotic exponent.

7.5 What these experiments do not establish

No finite corpus establishes universal checkpoint bounds, small presentations, or a success probability on arbitrary knots. No timeout is counted as an exact answer or as a completed baseline runtime. No exponent is estimated from the timing table. No synthetic word circuit is promoted to a knot input without group provenance. The tests and measurements support implementation claims and identify costs worth reducing; the asymptotic claims come from the explicit proofs.

8 Further research questions and a prioritized program

The most useful next step is not simply a larger benchmark. It is to separate structural hypotheses that might yield a quasi-polynomial theorem from engineering changes whose value must be established by controlled whole-query experiments. The questions below specify the desired object,

the proof obligation, and a useful first deliverable.

8.1 Structural questions bearing directly on the target

Q1. Which precisely defined residual class admits small presentation parameters?

After certified prime decomposition and the maintained structural, determinant, coloring, and Alexander tests, can the surviving class be shown to have a constructible circuit cut satisfying $(r + t) \log(d + 2) = O(\log^2 n)$? The filters must be fixed mathematically, not described as “all easy cases.” Begin with prime determinant-one or Alexander-trivial diagrams and identify whether large ordinary group rank persists. The displayed dihedral obstructions are removed by cheap coloring and therefore do not settle this residual question. A counterfamily would be as useful as a positive theorem, provided it survives the stated filters and includes a rank or cut lower bound.

Q2. Can one construct small presentations of unknots efficiently?

Every unknot group has a one-generator presentation, but that existence does not bound the cost of finding it from a difficult diagram. Seek a family of verified Tietze or geometric transformations with bounded move count, grammar growth, and compressed verification cost on unknots. Separate unknotted-input guarantees from the separate task of rejecting knotted inputs. A proof that a local search never stalls requires a global completeness argument, and any temporary word expansion must be charged in its actual representation.

Q3. Can the decision problem use smaller data than the whole group?

The rank obstructions rule out a universal small presentation of the entire group, not a small certificate tailored to triviality. One could seek verified decompositions, representation-preserving quotients, or local extension problems whose conjunction detects the unknot. A proposed quotient must preserve the direction needed: a nonabelian image of a quotient certifies knottedness, whereas failure to find one generally does not certify the unknot. A complete reduction needs both directions and a bound on the size and number of subproblems.

Q4. Can checkpoint gaps and frontier widths be controlled together?

Construct crossing orders or certified splicing moves with $g + w = O(\log^2 n)$ on a meaningful class of hard residual diagrams. The gap must include the crossing that triggers a replacement and the last segment; the width must be the actual temporary frontier. Study whether alternating reductions, arborescent decompositions, or controlled tangle sums give such orders after the already easy cases are removed. A bound only on final survivor count leaves the intermediate cost unresolved.

Q5. Can geometric hierarchy certificates supply the missing size theorem?

Translate the hierarchy parameters in [12] into explicit data structures and verified transitions. Bound hierarchy length, surface and pattern complexity, compressed cutting cost, and the bit length of every update. The displayed iteration bound $L(g + 1)^L$ is only one term in that analysis. A practical starting point is a small certificate checker for one cutting or simplification step, with independent topology reconstruction and a precise input-size theorem.

Q6. What exactly is established by the recent bridge-minimization claim?

Independently audit [17], including conversion from arbitrary input diagrams, completeness of the allowed moves, the application of Otal’s theorem to the unknot case, and polynomial bounds for their actual encoded execution. Record any lemma that requires an additional hypothesis and distinguish implementation tests from proof of that lemma. The present report’s results

do not assume the claim; resolving its status would materially change the broader research landscape.

8.2 Algebraic compilation and exact certificates

- Q7. How hard is the best word-circuit cut?** Study the optimization problem $\min_C(r + |C'|) \log(d_C + 2)$ on a word DAG. A first target is an exact dynamic program for trees, followed by a proved approximation or a parameterized algorithm for shared DAGs. Repeated subwords make local greedy cuts potentially suboptimal. Formal degree, actual monomial support, coefficient height, and compiler work can be tracked separately; an empirical support heuristic should not be substituted into the worst-case theorem without a bound.
- Q8. Can quaternion-specific identities improve the degree bound?** Repeated powers of a unit quaternion satisfy a quadratic recurrence, so Chebyshev-type representations may compress certain relators further. Any use of these identities must preserve all unit-quaternion solutions, including central generators, and count newly introduced scalar or quaternion variables. Determine whether power-heavy knot presentations admit smaller cuts than generic straight-line words. Avoid confusing a short recurrence with a constant-cost real formula.
- Q9. Can exact representation witnesses be checked independently?** The current optional solver records algebraic models and checks its assertions through the same exact engine. A stronger SAT artifact would give isolating intervals, defining polynomials, and independently verified sign tests for every equation and the noncommutativity inequality. For UNSAT, specify a checkable real-algebraic proof format or a verified decision backend. Establish verifier complexity in the certificate’s actual bit size; a compact approximate numerical model alone is insufficient.
- Q10. How should the known traceless-meridian theorem be used?** Kronheimer–Mrowka already prove complete detection with a meridian sent to i [9, Corollary 7.17]; this is not an open trace-existence question. Implement and independently validate Corollary 3.7, carrying explicit meridional provenance through generator changes. Compare its smaller complete formula with the current one-sided witness probe on the same residual corpus. A separate mathematical question concerns other prescribed trace values or trace schedules that improve real-algebraic decomposition. Such a restriction requires its own theorem, and generic nonmeridional generators must retain the unrestricted query.
- Q11. Can presentation provenance remain compressed across every move?** The repository already independently replays several compressed group certificates. Connect that verifier to the new circuit compiler so arbitrary generator changes switch correctly from the meridional inequality to the general commutator condition. Bound grammar size and assertion growth through long traces, not just individual word operations. The first deliverable should be a shared, versioned presentation object carrying origin, generator meanings, and exactly the current relators.

8.3 Earlier homological decisions and reusable geometry

- Q12. Can the flat-origin constraint remove the second rank safely?** For genuine minimal same-matching components, $\kappa = t$. Determine how to maintain a cheap connection or provenance certificate across scalar cancellation, direct-summand extraction, and total-rank

replacements. The last operation need not preserve genuine cube origin. A theorem covering exactly the executed scanner would permit using one rank where two are currently computed. Merely observing $t = \kappa$ in a corpus is not sufficient to install that shortcut.

- Q13. What constraints survive in higher jets or mixed matchings?** Lift the identity $\partial Q = \Gamma Q + Q \Gamma$ beyond scalar residue. Can it restrict higher-dot coefficients, transferred operations, or the shape of mixed-matching minimal blocks enough to bound multiplicities? The objective is a genuine realizability constraint on tangle-derived complexes. Arbitrary polynomial complexes over the dot algebra may be too broad to support the desired theorem, as the synthetic odd intervals already demonstrate.
- Q14. Can eligible components be maintained incrementally?** Current graph traversal and binary ranks can cost more than the scalar cancellations they avoid. Develop a dynamic index that certifies a whole one-matching direct summand and updates its scalar matrix after one crossing. It must detect all external attachments and survive transactional interruption. Evaluate saved cobordism compositions against the cost of component maintenance; the present easy replacement families regress.
- Q15. When should a scan precompute suffix responses?** Develop an adaptive policy that chooses independent setup, checkpoint responses, or full reverse compilation based on a proved cost estimate and observed demand. Aim for a competitive bound relative to the better strategy, including source geometry and abandoned work. A forward scanner may terminate before most precomputed suffixes are queried. The large all-stage timing ratios therefore cannot directly choose a default policy.
- Q16. Can a separator tree support local response updates?** Replace the linear order by a decomposition tree while retaining radical couplings and exact grounding. Prove a boundary contract that permits subtree changes without identifying eliminated interior vertices. Determine which snapshots and Euler summaries can be shared across primes or nearby orders despite exceptional pivot profiles. The established low-treewidth linear-algebra literature provides a starting point, not a substitute for the topology-specific gluing proof [5].

8.4 Evidence, dispatch, and a concrete sequence of deliverables

- Q17. Which residual corpus measures the intended improvement?** Assemble difficult unknots and knots that survive the exact existing front ends, with full PD provenance, mirrors, several orders, and duplicate detection. Include determinant-one and Alexander-trivial knots, inflated unknots, and examples with substantial repeated suffix interior work. A test label should come from an independent certificate or complete oracle. Synthetic word circuits remain useful for capacity tests but must be reported separately from knot-recognition instances.
- Q18. Which optional method improves complete queries reliably?** Run paired, shuffled whole-recognizer experiments with identical controls, fixed source hashes, explicit warmup rules, and all outcomes retained. Report CPU and wall time scopes, object and coefficient peaks, solver startup, model validation, and censored runs. Promotion to a default should require a reproducible benefit on the target residual corpus without a correctness or deadline regression. A small local improvement does not justify hiding a larger all-stage setup cost.

The proposed order is concrete. First, finish the residual corpus and attach independently checked

presentation provenance to the representation compiler. Next, reduce first-jet traversal overhead and prototype an adaptive response consumer, retaining the present backends as controls. In parallel, pursue either a circuit-cut theorem for a precisely filtered class or a geometric bound on checkpoint gaps and frontier width. A successful structural theorem would immediately plug into Corollary 6.1; until then, the optional implementations and recorded negative results provide a disciplined experimental platform.

A Reproduction and integration notes

A.1 Pinned source and package contents

The baseline is the public repository commit `1be2abc8cc743c1a3b5cb7fdfeb650697f8e2d5b`. The companion archive includes this article’s complete L^AT_EX source and bibliography, the compiled PDF, an integration patch, a reviewable tree of changed and new repository files, and a runnable source snapshot with the independent test-oracle files needed by the existing suite. Machine-readable manifests record source provenance and SHA-256 digests. Temporary bytecode caches, compiler intermediates, and failed retrieval artifacts are excluded.

The new implementation files are `fastunknot/su2.py`, `fastunknot/first_jet.py`, `fastunknot/first_jet_scan.py`, and `fastunknot/streaming_response.py`. Each has focused tests. Raw measurements and inputs live in `fast/results/`; replay scripts are grouped by experiment. The default recognizer does not automatically select these new research backends.

One existing file, `fastunknot/normal_surface.py`, receives an inherited correction from the preceding research continuation. Repeated `communicate(None)` calls after a timeout can lose an incompletely sent input payload on the supported Python subprocess path. The corrected worker performs exactly one `communicate(payload)` in a thread, polls its completion under the cooperative deadline, and kills, reaps, and joins on interruption. This resolves the existing `test_communication_retains_input_across_polls` regression. It is an integration fix with prior provenance, not a new mathematical or recognition-performance result of this article.

A.2 Build and test commands

The article uses standard L^AT_EX packages; no external network source is needed to compile it. From the article directory:

```
latexmk -pdf -interaction=nonstopmode -halt-on-error \
unknot_presentations_and_early_certificates.tex
```

An already generated bibliography is included for installations that prefer a plain `pdflatex` build; a complete refresh uses BibTeX through `latexmk`. The archived PDF is rendered and visually checked after compilation.

From the bundled or patched `Topology/UnknotRecognition/fast` directory:

```
python3 -m unittest discover -s tests -q
python3 first_jet_research/audit.py --fast-root . \
  --output results/first_jet_discovery_local.json
python3 benchmark_first_jet.py \
  --output results/first_jet_benchmark_local.json
python3 response_research/audit_streaming_response.py \
  --output results/streaming_response_audit_local.json
python3 response_research/benchmark_streaming_response.py \
  --output results/streaming_response_local.json \
  --sizes 16 32 64 128 --rounds 7 --large-stream 2048
python3 response_research/benchmark_streaming_response.py \
  --output results/streaming_response_large_local.json \
  --sizes 256 --rounds 3 --large-stream 8192
python3 su2_research/audit_su2.py \
  --output results/su2_audit_local.json
python3 su2_research/benchmark_su2.py \
  --output results/su2_benchmark_local.json
```

The detailed README gives the accepted output-path and size flags of each script. The recorded optional real solver is `z3-solver==5.1.0.0`, reporting core version 5.1.0. The symbolic compiler, exact rational-witness tests, finite-group checks, first-jet code, and response code do not need that dependency. Solver-dependent tests skip when it is absent.

The complete integrated suite depends on the bundled independent reference modules from reports 24, 26, and 28. They are unchanged copies of pinned repository files, not a second implementation of the new incremental state inside its oracle. Existing optional Regina tests remain conditional on that external dependency.

A.3 Example interfaces

The following generic query has no knot verdict attached:

```
from fastunknot.su2 import compile_presentation, solve_formula

# The trefoil presentation aba = bab, supplied here only as a group.
formula = compile_presentation(
    [1, 2], [(1, 2, 1, -2, -1, -2)], chunk_size=3
)
print(formula.summary())
print(solve_formula(formula, seconds=1.0))
```

Use the validated diagram constructor to obtain a knot decision:

```
from fastunknot.diagram import Diagram
from fastunknot.su2 import su2_decide

d = Diagram.from_braid(3, [1, -2, 1, -2])
answer = su2_decide(d, presentation="tietze", seconds=1.0,
                    trace_zero_probe=True)
print(answer["status"])
```

The trace-zero flag enables a one-sided witness probe only. This interface still queries the full formula after a failed restricted probe; it does not exploit the stronger complete specialization in Corollary 3.7. Export an exact SMT-LIB input with the standalone CLI's `-smt2` option.

The rank-only scanner accepts validated PD data and an optional supplied order:

```
from fastunknot.first_jet_scan import first_jet_khovanov_decide

answer = first_jet_khovanov_decide(
    d.pd, seconds=2.0, max_objects=50000,
    first_jet_max_work=1000000
)
print(answer["status"], answer["first_jet_stats"])
```

Its global scan limit is an exception in this raw API, consistent with the maintained raw scanner contract. A caller must handle that as unresolved. It must not expose replacement degree counts as the original graded homology.

Finally, the response compiler takes a crossing order and a prime:

```
from fastunknot.streaming_response import compile_suffix_responses

responses = compile_suffix_responses(
    d.pd, order=list(range(d.crossings)), prime=101,
    max_boundary=100, max_stored_elements=1000000
)
# Query a stage only with a checked matching on that stage's boundary.
```

The storage allowance counts retained matrix entries; it is not a hard process-memory cap. Source data and a current temporary matrix have the additional costs proved in the response theorem.

A.4 Reproducibility and trust limitations

All mathematical proofs are presented for review rather than described as formally verified. Independent implementations and cross-checks reduce implementation risk, but finite tests do not replace those proofs. The optional real solver is an explicit trusted backend. SAT models are retained as exact algebraic expressions and evaluated by that same backend, while UNSAT is its exact decision result; no independent UNSAT proof checker is bundled.

The source snapshot is a convenience for reproduction, not a substitute for reviewing the small integration patch against the pinned repository. The archive is prepared for later repository integration. It does not represent a pushed commit or a published change to ProveIt.

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